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CRB3 navigates Rab11 trafficking vesicles to promote γTuRC assembly during ciliogenesis

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Abstract

The primary cilium plays important roles in regulating cell differentiation, signal transduction, and tissue organization. Dysfunction of the primary cilium can lead to ciliopathies and cancer. The formation and organization of the primary cilium are highly associated with cell polarity proteins, such as the apical polarity protein CRB3. However, the molecular mechanisms by which CRB3 regulates ciliogenesis and the location of CRB3 remain unknown. Here, we show that CRB3, as a navigator, regulates vesicle trafficking in γ-TuRC assembly during ciliogenesis and cilium-related Hh and Wnt signaling pathways in tumorigenesis. *Crb3* knockout mice display severe defects of the primary cilium in the mammary ductal lumen and renal tubule, while mammary epithelial-specific *Crb3* knockout mice exhibit promotion of ductal epithelial hyperplasia and tumorigenesis. CRB3 is essential for lumen formation and ciliary assembly in the mammary epithelium. We demonstrate that CRB3 localizes to the basal body and that CRB3 trafficking is mediated by Rab11-positive endosomes. Significantly, CRB3 interacts with Rab11 to navigate GCP6/Rab11 trafficking vesicles to CEP290, resulting in intact γ-TuRC assembly. In addition, CRB3-depleted cells are unresponsive to the activation of the Hh signaling pathway, while CRB3 regulates the Wnt signaling pathway. Therefore, our studies reveal the molecular mechanisms by which CRB3 recognizes Rab11-positive endosomes to navigate apical vesicle trafficking in effective ciliogenesis, maintaining cellular homeostasis and tumorigenesis.

eLife assessment

This study is **useful** to scientists studying cell polarity, epithelial morphogenesis, cancer, and primary cilia. The authors confirm the need for CRB3A/B in regulating ciliogenesis by using a combination of a mammary epithelial cell-specific conditional Crb3 knockout mouse model, and cellular, molecular and biochemical approaches. The results are **solid**, supporting and extending previous findings; the results also indicate that CRB3 affects ciliogenesis by a still **incompletely** understood mechanism involving Rab11 and gamma-TuRC.

Introduction

Epithelial tissues are the most common and widely distributed tissues in the body, and they perform several particular functions. The integrity of the cellular architecture composed of epithelial cells is necessary for these functions. Establishing apical-basal cell polarity within epithelial cells is a crucial biological process for epithelial homeostasis. Unfortunately, the loss of apical-basal cell polarity is one of the most significant hallmarks of advanced malignant tumors[1], and several reports have shown that the loss of some polarity proteins causes disordered cell polarity within epithelial cells, which cannot tightly control cellular growth and migration, ultimately leading to tumor formation, progression, and metastasis[2, 3]. However, the relationship between polarity proteins and tumorigenesis remains incomplete.

The Crumb complex, located apically, is vital in regulating and maintaining apical-basal cell polarity within epithelial cells. In mammals, three Crumbs orthologs are CRB1, CRB2, and CRB3, and only CRB3 is widely expressed in epithelial cells[4]. Our previous studies have shown that CRB3 is expressed at low levels in renal and breast cancers, and abnormal CRB3 expression leads to the disrupted organization of MCF10A cells in three-dimensional (3D) cultures[5–7]. Inhibition of CRB3 impairs contact inhibition and leads to migration, invasion and tumorigenesis of cancer cells[6, 8]. Contact inhibition results in growth arrest and causes epithelial cells to enter the quiescent phase, inducing primary cilium formation.

The primary cilium, an antenna-like microtubule-based organelle, is a sensorial antenna that regulates cell differentiation, proliferation, polarity, and tissue organization in most types of mammalian cells[9]. Dysfunction of the primary cilium can lead to developmental and degenerative disorders called ciliopathies, such as polycystic kidney disease, nephronophthisis, retinitis pigmentosa, and Meckel syndrome[10]. Remarkably, many cancers, including melanoma and breast, renal, pancreatic, and prostate cancer, exhibit loss of the primary cilium, most likely during the early stages of tumorigenesis[11–15]. The primary cilium is a significant extension on the apical surfaces of epithelial cells and is maintained by polarized vesicular traffic[16]. Studies have shown that CRB3 localizes in the primary cilium and that depletion of CRB3 leads to primary cilium loss in Madin-Darby canine kidney (MDCK) cells[17, 18], indicating that CRB3 is necessary for the initiation of ciliogenesis. However, how CRB3 promotes ciliogenesis on the apical surfaces of epithelial cells remains unknown.

Here, we describe a novel conditional deletion of *Crb3* in mice and show that Crb3 is required to assemble the primary cilium in the mammary gland, kidney and MEF cells from *Crb3* knockout mice and that *Crb3* deletion promotes breast cancer progression *in vivo*. Additionally, we found that CRB3 interacts with Rab11 and navigates GCP6/Rab11 trafficking vesicles to the basal body of the primary cilium in mammary epithelial cells. Furthermore, we identified that CRB3 regulates the ciliary Hedgehog (Hh) and Wnt signaling pathways in breast cancer. Thus, there is unexpected diversity in the polarity protein CRB3 navigating polarized Rab11-dependent traffic, leading to a novel assembly mechanism of the γ-tubulin ring complex (γTuRC) in ciliogenesis.

Results

Crb3 deletion mice exhibit smaller size and anophthalmia

Since *Crb3* knockout mice die shortly after birth[19], we generated a novel transgenic mouse model with conditional deletion of *Crb3* using the Cre-loxP system. The loxP sites flanked either side of exon 3 in the *Crb3* gene (Supplementary Fig. 1A). In mammals, CRB3 has two isoforms, CRB3a and CRB3b, distinguished by alternative splicing within the fourth exon of the *CRB3* gene, which in turn produces a protein with 23 amino acid differences at the C terminus[17]. According to the construction strategy, we are knocking out both isoforms *Crb3a* and *Crb3b* in the mice. The positive embryonic stem (ES) cell clone was confirmed for the wild-type gene (*Crb3*$^{wt/wt}$) and recombinant allele (*Crb3*$^{wt/fl}$) by Southern blotting (Supplementary Fig. 1B). The genotypes of the wild type, heterozygotes and homozygotes (*Crb3*$^{fl/fl}$) were detected with specific primers (Supplementary Fig. 1C). We first intercrossed *Crb3*$^{fl/fl}$ with CMV enhancer/chicken β-actin promoter (CAG)-Cre mice and found that *Crb3*$^{fl/fl}$, *Crb3*$^{wt/fl}$; CAG-Cre and *Crb3*$^{fl/fl}$; CAG-Cre pups were born normally and well developed in size. However, most *Crb3*$^{fl/fl}$; CAG-*Cre* pups died within a few days after birth. Few surviving *Crb3*$^{fl/fl}$; CAG-*Cre* mice were smaller and showed anophthalmia than littermate *Crb3*$^{fl/fl}$ mice at 4 weeks old (**Fig. 1A-B**). The eye is an organ with perfect apical-basal cell polarity. However, *Crb3* knockout mice showed ocular abnormalities. Together, these results indicate that Crb3 is necessary for eye development.

Deletion of mammary epithelial-specific *Crb3* causes ductal epithelial hyperplasia and tumorigenesis

Based on our earlier studies, which showed that inhibiting CRB3 impairs the organization of breast epithelium, we set out to directly explore the role of Crb3 in mammary gland development. We first developed epithelial cell-specific deletion of *Crb3* in the mammary gland by intercrossing *Crb3*$^{fl/fl}$ mice with mouse mammary tumor virus long terminal repeat promoter (MMTV)-*Cre* mice. We confirmed the expression of Crb3 in the mammary gland using immunoblotting (both isoforms Crb3a and Crb3b), real-time quantitative PCR, and immunohistochemistry, and Crb3 expression was deficient in mammary epithelial cells with Crb3 knockout (Supplementary Fig. 1D-F).

To assess the function of Crb3 in the development of mammary epithelial cells, we analyzed whole mammary mounts from 8-week-old virgin *Crb3*$^{fl/fl}$;MMTV-*Cre* and littermate *Crb3*$^{fl/fl}$ mice. *Crb3*$^{fl/fl}$;MMTV-*Cre* mice displayed significantly increased numbers of terminal end buds (TEBs) and bifurcated TEBs compared to *Crb3*$^{fl/fl}$ mice (**Fig. 1 C-E**). Histopathology results revealed that normal ductal epithelial cells were arranged as a monolayer in *Crb3*$^{fl/fl}$ mice, while *Crb3* knockout led to ductal epithelial hyperplasia with increased ductal thickness in *Crb3*$^{fl/fl}$;MMTV-*Cre* mice (**Fig. 1 F**).

To investigate the role of Crb3 in breast cancer tumorigenesis, we observed mammary glands from 9-week-old virgin polyomavirus middle T antigen (PyMT)-cKO-Crb3 and PyMT-WT mice. The mammary glands of PyMT mice progressed to early carcinomas and lost their normal epithelial architecture[20]. The PyMT mouse model is widely used to study tumor initiation and development, with a histological progression similar to that of human breast cancer. PyMT-WT mice exhibited loss of epithelial features and developed breast cancer, while PyMT-cKO-Crb3 mice

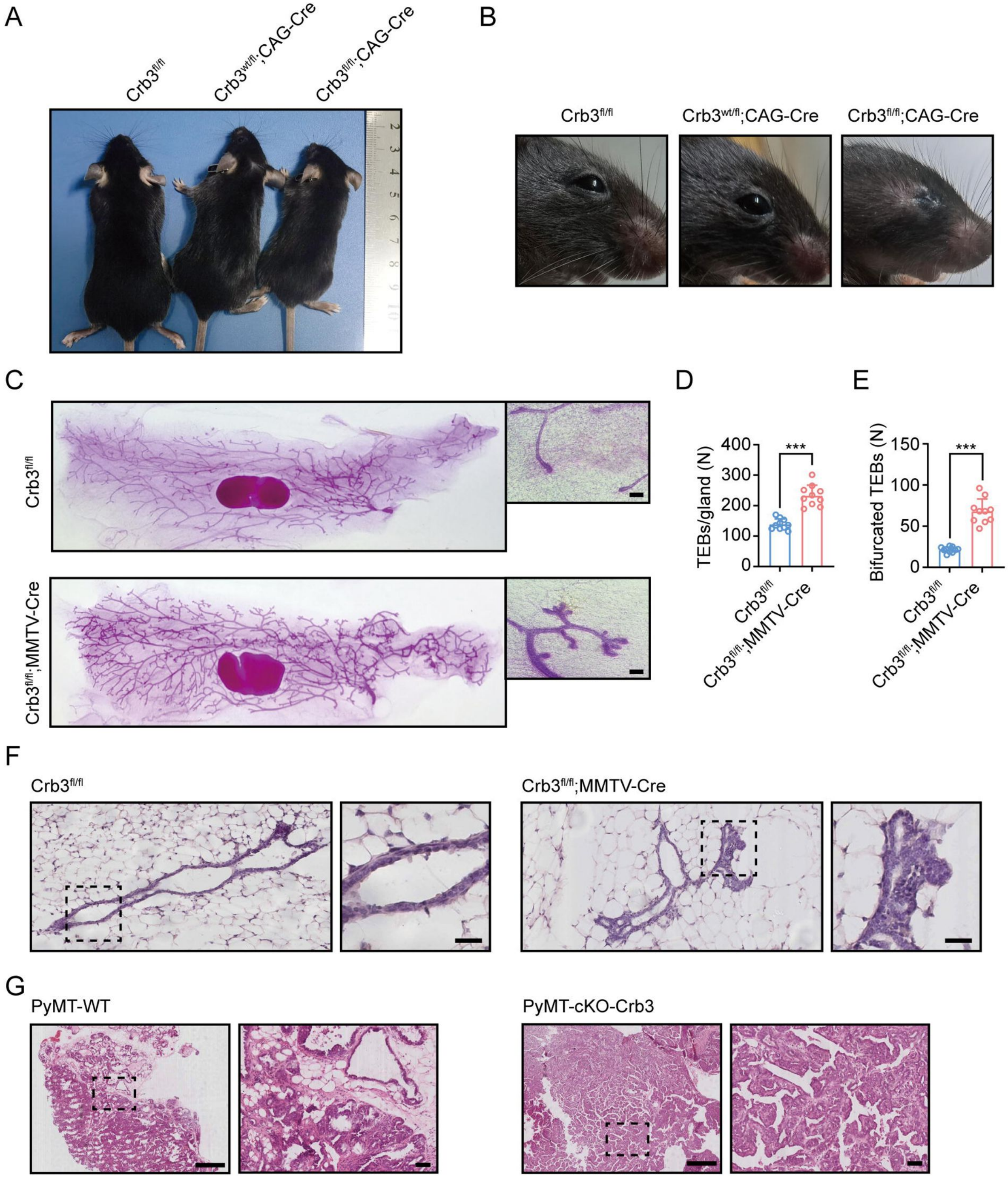

Figure 1.

Crb3 knockout mice exhibit smaller sizes and ocular abnormalities, and mammary epithelial cell-specific *Crb3* knockout leads to ductal epithelial hyperplasia and promotes tumorigenesis.

A, B. Representative whole bodies (A) and eyes (B) from littermate *Crb3*$^{fl/fl}$, *Crb3*$^{wt/fl}$;CAG-*Cre* and *Crb3*$^{fl/fl}$;CAG-*Cre* mice at 4 weeks old. C. Representative mammary whole mounts from littermate *Crb3*$^{fl/fl}$ and *Crb3*$^{fl/fl}$;MMTV-*Cre* mice at 8 weeks old with Carmine-alum staining. (scale bars, 200 μm) D, E. Quantification of the average number of TEBs and bifurcated TEBs in littermate *Crb3*$^{fl/fl}$ (n=10) and *Crb3*$^{fl/fl}$;MMTV-*Cre* (n=10) mice at 8 weeks old. F. Representative images of mammary glands in littermate *Crb3*$^{fl/fl}$ and *Crb3*$^{fl/fl}$;MMTV-*Cre* mice stained with H&E. (scale bars, 50 μm) G. Representative images of primary tumors stained with H&E in PyMT-WT and PyMT-cKO-*Crb3* mice at 9 weeks old. (scale bars, left 500 μm, right 50 μm) The magnified areas of boxed sections are shown in the right panels. Bars represent the means ± SD; Unpaired Student's *t* test, ***P<0.001.

developed larger tumors with faster tumor progression and a poorly differentiated phenotype (**Fig. 1G**). Thus, these results suggest that Crb3 knockout leads to ductal epithelial hyperplasia and promotes tumorigenesis.

CRB3 knockdown inhibits lumen formation in acini of mammary epithelial cells

MCF10A cells are spontaneously immortalized human breast epithelial cells with the characteristics of normal breast epithelium[21]. In 3D basement membrane culture conditions, MCF-10A cells form acini similar to mammary epithelial acini. This makes them a valuable system for studying the morphogenesis and oncogenesis of the mammary epithelium[22]. To confirm the function of CRB3 in mammary epithelial acini formation, we first knocked down the expression of both CRB3a and CRB3b isoforms using lentiviral shRNA in MCF10A cells (Supplementary Fig. 2A and B). Under the 3D culture system, we found that MCF10A cells infected with a non-targeting shRNA lentivirus formed acini after 6 days (D6) and eventually formed polarized acini with a hollow lumen at D14 (**Fig. 2A**). However, after CRB3 knockdown, MCF10A cells formed more and larger aberrant acini without a lumen after D6 (**Fig. 2A** and **B**). Cell proliferation assays of MCF10A cells showed that CRB3 knockdown increased the proliferation rate and accelerated G1 to S phase progression (Supplementary Fig. 2C-E).

To investigate the reason for aberrant acini without lumens, we further studied apoptosis and mitotic spindle orientation during lumen formation in a 3D culture system. The immunofluorescence (IF) results showed that more internal cleaved-caspase 3-positive acini were observed in the control groups than in the CRB3 knockdown groups at D9 (**Fig. 2C** and **D**). Moreover, CRB3b overexpression significantly promoted apoptosis in breast cancer cells (Supplementary Fig. 2F-H). Similarly, the staining of α-tubulin in mitotic cells demonstrated more misorientation of the mitotic spindle in CRB3 knockdown groups (**Fig. 2E** and **F**).

In addition to verifying the effect of *Crb3* deletion on promoting proliferation and inhibiting apoptosis *in vivo*, we detected these markers in primary tumor tissue from PyMT-WT and PyMT-cKO-Crb3 mice. *Crb3* deletion increased the numbers of proliferative and mitotic cells, and decreased the numbers of apoptotic cells in primary tumors (**Fig 2G** and **H**). These results demonstrate that CRB3 could promote the proliferation of mammary epithelial cells, disrupt the mitotic spindle orientation, and protect the internal cells from apoptosis during acini formation, ultimately leading to irregular lumen formation and tumor progression.

CRB3 regulates primary cilium formation

The primary cilium is an apical antenna-like extensions in epithelial cells that is known to play an important role in controlling lumen formation[23]. Previous studies have reported that CRB3 knockdown leads to primary cilium loss in MDCK cells[17, 18]. However, the molecular mechanism that regulates ciliogenesis on apical surfaces is still unclear. To investigate the altered effect of CRB3 on ciliogenesis both *in vivo* and *in vitro*, we examined the formation of primary cilia in control MCF10A cells and CRB3 knockdown cells. We found that control MCF10A cells formed primary cilia after becoming confluent, while CRB3 knockdown led to significant ciliogenesis defects (**Fig. 3A**). The number of cells with primary cilium was significantly reduced compared to the control group, but the length of the primary cilium that did exist was not different (**Fig. 3B**). Conversely, CRB3b conditional overexpression by adding doxycycline (+dox) resulted in restoring ciliary assembly in MCF7 cells (**Fig. 3C**, Supplementary Fig. 3A). CRB3b overexpression increased the proportion of primary cilium formation in breast cancer cells, and the length of the restored primary cilium was increased (**Fig. 3D**). Importantly, the primary cilium could be directly visualized using scanning electron microscopy (SEM) (**Fig. 3E**). After CRB3 knockdown, the number of MCF10A cells with primary cilia was significantly decreased, while CRB3b conditional overexpression did not increase these populations (**Fig. 3F**).

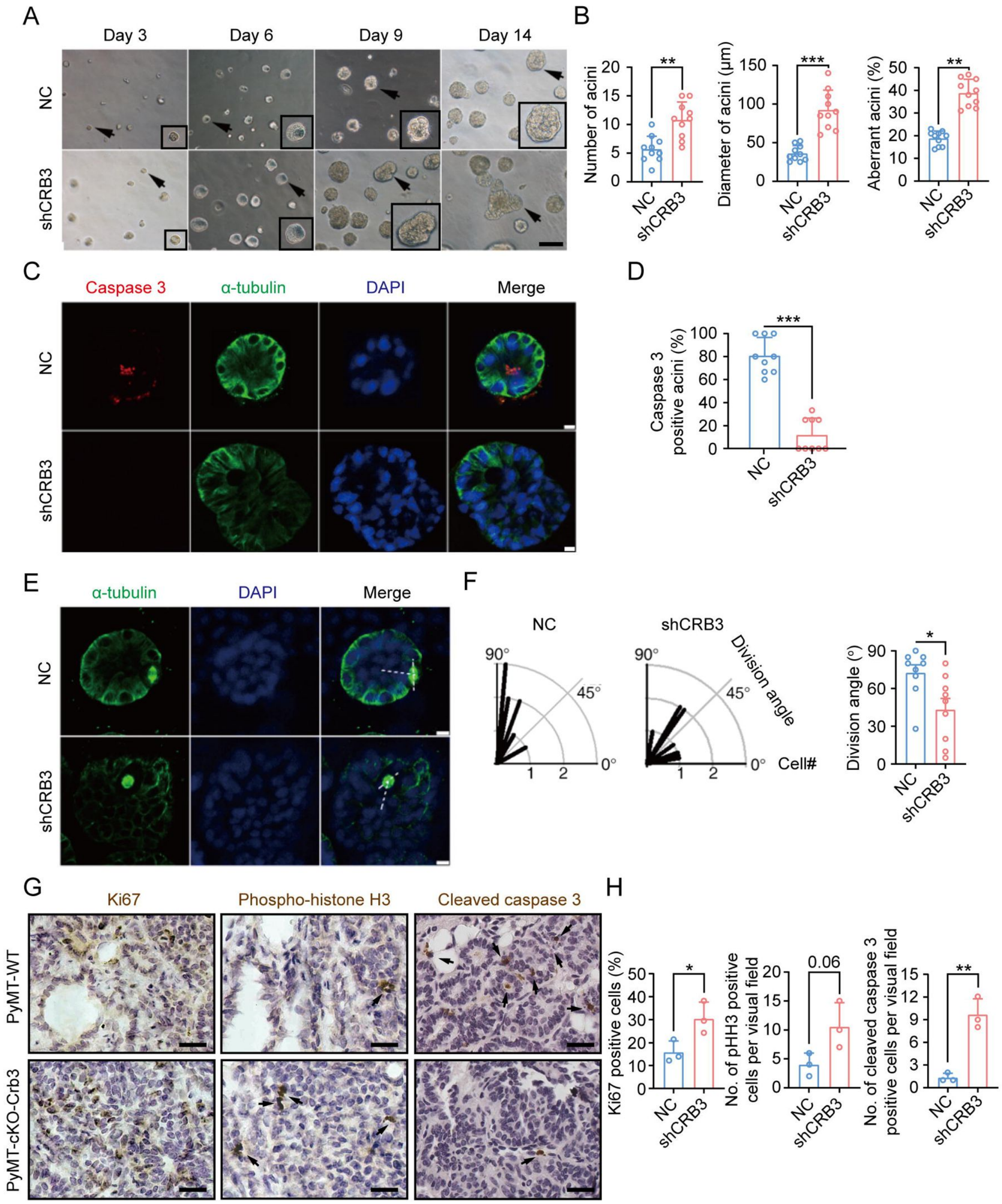

Figure 2.

CRB3 knockdown inhibits acinar formation of mammary epithelial cells in a 3D culture system.

A. Representative effect of CRB3 on acinar formation in the 3D culture system at days 3, 6, 9 and 14. (The magnified areas of the marked arrows are shown in the lower right corner) B. Quantification of the average number, diameter and aberration of acini (At least 3-4 random micrographs were analyzed for each condition in each of three independent experiments, n=10). C. Immunofluorescence showing apoptosis during lumen formation. Caspase 3 (red), α-tubulin (green), and DNA (blue). D. Quantification of the proportion of caspase 3 positive acini (At least 3-4 random micrographs were analyzed for each condition in each of three independent experiments). E. Immunofluorescence showing the mitotic spindle orientation during lumen formation. α-tubulin (green) and DNA (blue). F. Quantification of division angle (At least 3-4 acini were randomly examined for each condition in each of three independent experiments). G. Immunohistochemical analyses of Ki67, phospho-histone H3 and cleaved caspase 3 in primary tumors from PyMT-WT (n=3) and PyMT-cKO-*Crb3* (n=3) mice at 9 weeks old. (Positive cells are marked by arrows; scale bars, 25 μm) H. Quantification of the number of Ki67, phospho-histone H3 and cleaved caspase 3 positive cells from IHC staining images. Bars represent means ± SD; Unpaired Student's *t* test, * *P*<0.5, ** *P*<0.01, ****P*<0.001.

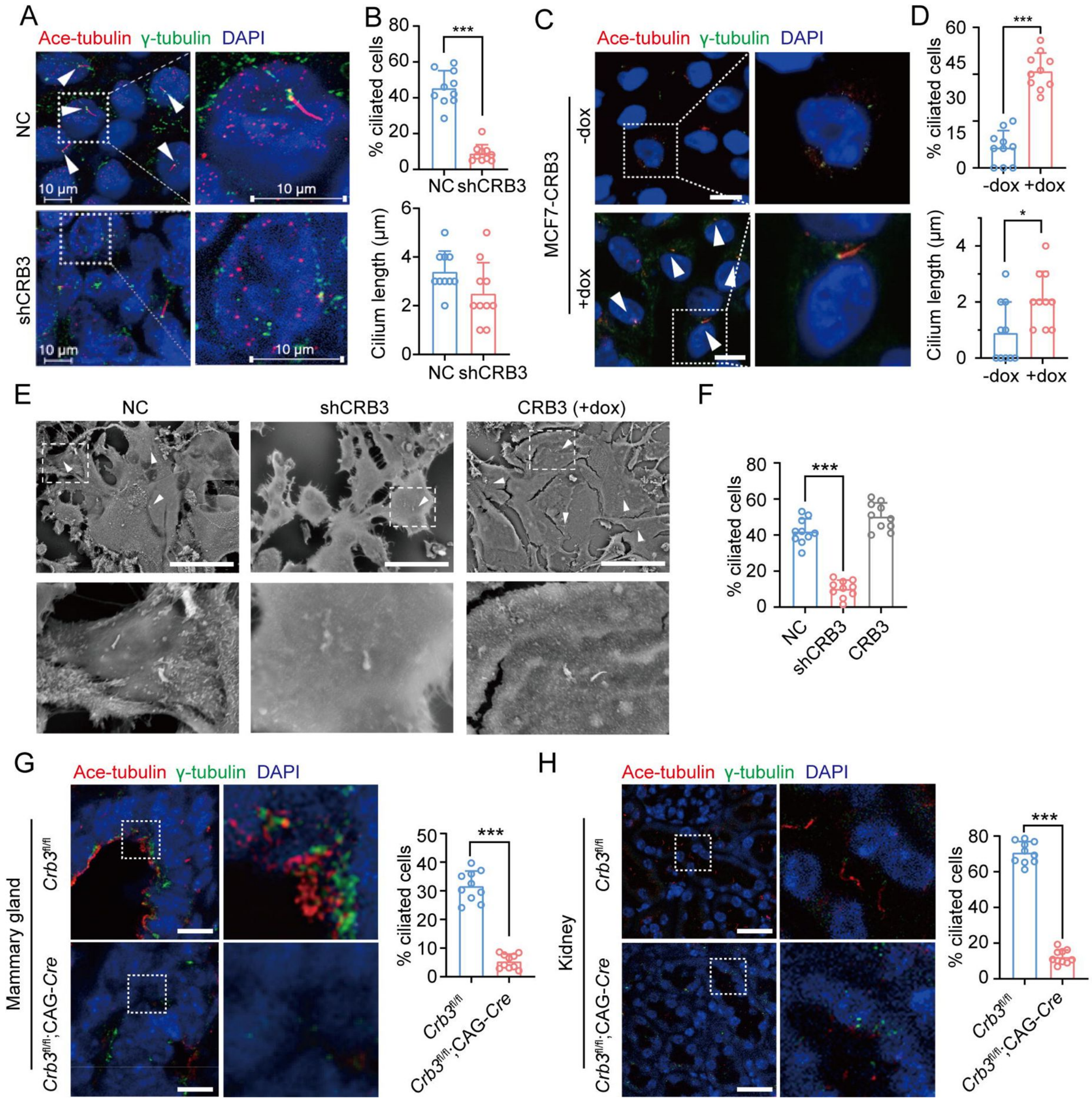

Figure 3.

CRB3 alters primary cilium formation in mammary cells, mammary ductal lumen and renal tubule from *Crb3*$^{fl/fl}$;CAG-*Cre* mice. A. C. Representative images of immunofluorescent staining of primary cilium formation with CRB3 knockdown in MCF10A cells and CRB3b conditional overexpression upon dox induction in MCF7 cells. Acetylated tubulin (red), γ-tubulin (green), and DNA (blue). (The primary cilium is marked by arrows; scale bars, 10 μm) B. D. Quantification of the proportion of cells with primary cilium formation and the length of the primary cilium (At least 3-4 random micrographs were analyzed for each condition in each of three independent experiments, n=10). E. Representative scanning electron microscope images of primary cilium formation with CRB3 knockdown and CRB3b conditional overexpression in MCF10A cells. (The primary cilium is marked by arrows; scale bars, 50 μm) F. Quantification of the proportion of cells with primary cilium formation (At least 3-4 random micrographs were analyzed for each condition in each of three independent experiments, n=10). G. H. Representative immunofluorescent staining of primary cilium formation in the mammary ductal lumen and renal tubule from *Crb3*$^{fl/fl}$ (n=10) and *Crb3*$^{fl/fl}$;CAG-*Cre* (n=10) mice, respectively. Acetylated tubulin (red), γ-tubulin (green), and DNA (blue). (scale bars, 25 μm) Bars represent the means ± SD; Unpaired Student's *t* test, **P*<0.05, ****P*<0.001.

To further support these findings *in vivo*, we used breast and kidney tissue to assess ciliogenesis in the *Crb3* knockout mouse model. Renal epithelial cells can form prominent primary cilia, and the absence of primary cilia leads to ciliopathies. We used immunofluorescence to detect ciliary formation in the mammary ductal lumen and renal tubule. Compared to tissues of *Crb3*$^{fl/fl}$ mice, *Crb3* knockout led to the absence of the primary cilium in the mammary ductal lumen and renal tubule from *Crb3*$^{fl/fl}$; CAG-*Cre* mice (**Fig. 3G, H**). Specifically, the proportion of cells forming the primary cilium plummeted (**Fig. 3G, H**). This same phenomenon was also observed in mouse embryonic fibroblasts (MEFs) from *Crb3*$^{fl/fl}$; CAG-*Cre* mice (Supplementary Fig. 3B, C). We conclude that CRB3 plays a central role in ciliogenesis in various tissues and mammary cells.

CRB3 localizes to the basal body of the primary cilium

CRB3, a polarity protein, is typically localized on the tight junctions of the apical epithelium membrane[24]. Additionally, we noted that CRB3b is found to be closely colocalized with centrosomes at prophase, metaphase, and anaphase[17]. To investigate the relationship between CRB3 and ciliary formation, we first examined CRB3 localization in MCF10A cells expressing CRB3-GFP. In MCF10A cells, exogenous CRB3b was mainly localized at cell tight junctions and could colocalize with pericentrin (a centrosome marker) (**Fig. 4A**). And approximately 40% of cells showed this co-localization of exogenous CRB3b and pericentrin (**Fig. 4B**). We verified this phenomenon by observing the localization of endogenous CRB3 (both isoforms CRB3a and CRB3b) with another centrosome marker. Similarly, we detected that CRB3 accumulated on one side of the cytoplasm and had a CRB3 foci located at the basal body represented by γ-tubulin in MCF10A cells (**Fig. 4C, D**). Importantly, CRB3 knockdown disturbed the accumulation of γ-tubulin at the basal body in confluent quiescence cells (**Fig. 4C, E**). To demonstrate the relationship of this colocalization to the primary cilium, we employed acetylated tubulin to mark the primary cilium. Double immunostaining showed that this CRB3 foci was the basal body of the primary cilium, and CRB3 knockdown inhibited ciliary assembly in MCF10A cells (**Fig. 4F, G**). These results suggest that CRB3 is located at the basal body of the primary cilium and that CRB3 knockdown disturbs the accumulation of γ-tubulin in quiescent cells.

CRB3 trafficking is mediated by Rab11-positive endosomes

To better investigate the molecular mechanisms of CRB3 in regulating primary cilium formation, we examined some important genes related to ciliogenesis. However, CRB3 knockdown did not affect the mRNA expression of these genes in MCF10A cells (Supplementary Fig. 4A). Then, we performed coimmunoprecipitation (co-IP) tandem mass spectrometry using tagged exogenous CRB3b as bait to identify its interacting proteins (Supplementary Fig. 4B). Pathway aggregation analysis of these proteins revealed that these interacting proteins of CRB3b were significantly involved in Golgi vesicle transport and vesicle organization (**Fig. 5A**). We focused on these pathways because smaller distal appendage vesicle (DAV) formation is critical for ciliogenesis initiation, and it requires the Rab GTPase Rab11-Rab8 cascade to function[25]. Interestingly, some Rab small GTPase family members identified as CRB3b-binding proteins were aggregated in these pathways, such as Rab10, Rab11A, Rab11B, Rab14, Rab1A, Rab1B, Rab21, Rab2A, Rab32, Rab35, Rab38, Rab5B, Rab5C, and Rab6A. Several centriolar proteins were also identified, including tubulin gamma-1, tubulin gamma-2, CENP-E, CEP290, CEP192, CEP295, and CEP162 (**Fig. 5B**).

Rab11-positive vesicles bind to centrosomal Rabin8, leading to the formation of ciliary membrane[26]. This finding suggested that the trafficking of Rab11-positive vesicles and polarized vesicles is essential for early ciliary assembly. Next, we examined the polarized vesicle traffic of CRB3 in MCF10A cells. Our studies found that CRB3 could colocalize with EEA1-positive early endosomes, CD63-positive late endosomes and Rab11-positive recycling endosomes (**Fig. 5C**). In addition, after treatment with dynasore, an endocytosis inhibitor, CRB3 significantly accumulated at the cell membrane after 2 hours, and the colocalization of CRB3 with EEA1-, CD63-,

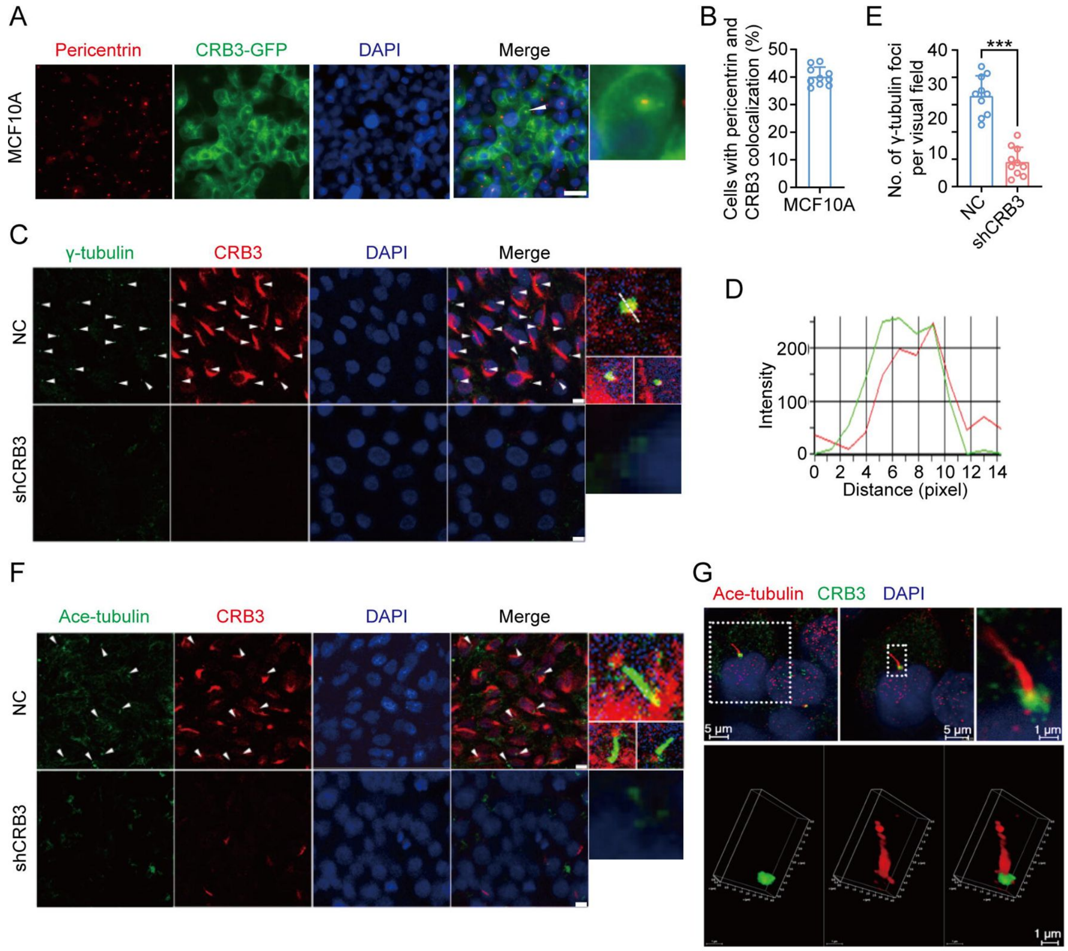

Figure 4.

CRB3 localizes to the basal body of the primary cilium.

A. Immunofluorescence showing the colocalization of exogenous CRB3b with centrosomes in MCF10A cells. Pericentrin, a marker of centrosome (red), CRB3-GFP (green), and DNA (blue). (Colocalization is marked by arrows; scale bars, 10 μm) B. Quantification of the proportion of cells with pericentrin and exogenous CRB3b colocalization (At least 3-4 random micrographs were analyzed for each condition in each of three independent experiments, n=10). C. Another colocalization of endogenous CRB3 with the basal body in MCF10A cells. γ-tubulin is a marker of the centrosome and basal body of the primary cilium (green), CRB3 (red), and DNA (blue). (Colocalization is marked by arrows; scale bars, 10 μm) D. Corresponding fluorescence intensity profile across a section of the array, as indicated by the dashed white line in (C). E. Quantification of the number of γ-tubulin foci in MCF10A cells from IF staining images (At least 3-4 random micrographs were analyzed for each condition in each of three independent experiments, n=10). F. Double immunostaining displaying the colocalization of CRB3 with the primary cilium in MCF10A cells. Acetylated tubulin (green), CRB3 (red), and DNA (blue). (Colocalization is marked by arrows; scale bars, 10 μm) G. Fluorescence 3D reconstruction of CRB3 and primary cilium colocalization. Acetylated tubulin (red), CRB3 (green), and DNA (blue). Bars represent the means ± SD; Unpaired Student's *t* test, ***P<0.001.

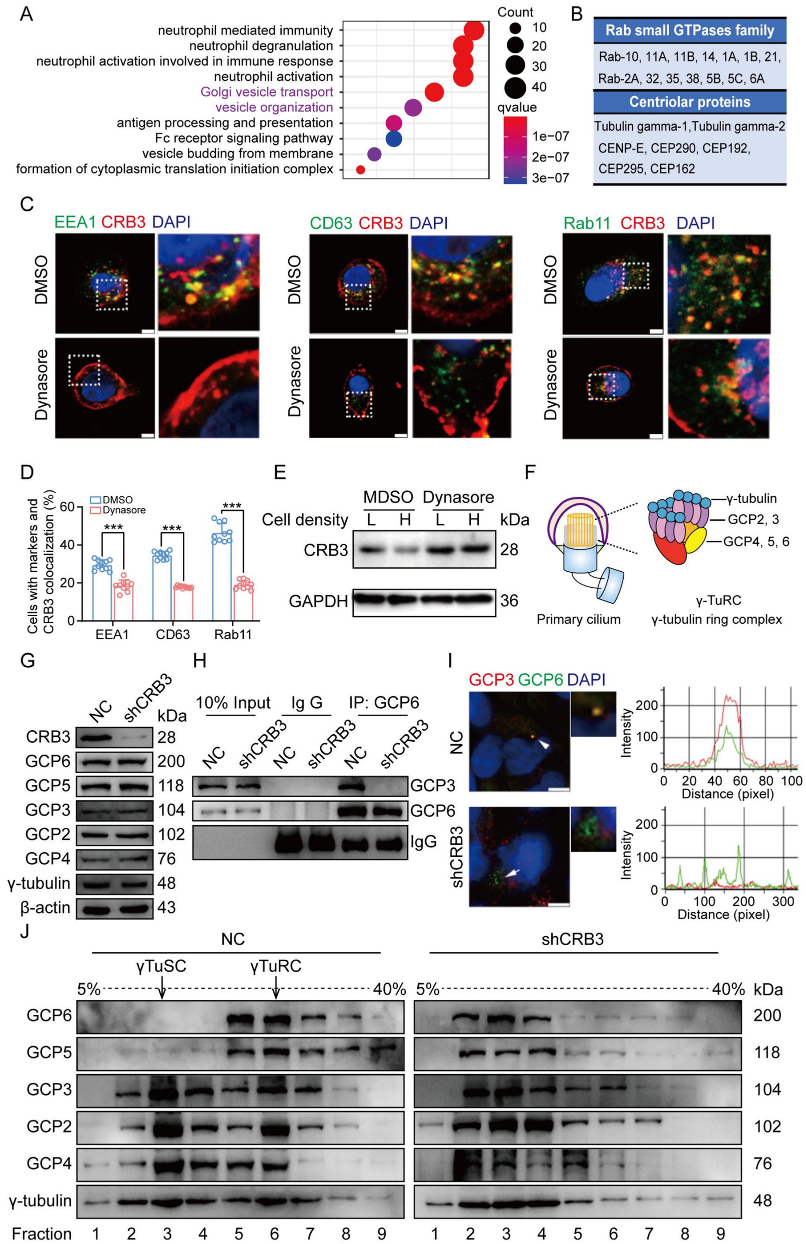

Figure 5.

CRB3 trafficking is mediated by Rab11-positive endosomes, and CRB3 knockdown destabilizes γTuRC assembly during ciliogenesis.

A. Pathway aggregation analysis of CRB3b binding proteins identified by mass spectrometry in MCF10A cells. B. Table of some Rab small GTPase family members and centriolar proteins identified as CRB3b binding proteins. C. Immunofluorescence showing the colocalization of CRB3 with EEA1-, CD63-, and Rab11-positive endosomes in MCF10A cells. EEA1, CD63, Rab11 (green), CRB3 (red), and DNA (blue). (scale bars, 10 μm) D. Quantification of the proportion of cells with these markers and CRB3 colocalization (n=10). E. Western blotting showing the levels of CRB3 in MCF10A cells treated with dynasore at different cell densities. F. The structure diagram of γTuRC (γ-tubulin ring complex). G. Immunoblot analysis of the effect of CRB3 on γTuRC molecules in MCF10A cells. H. Coimmunoprecipitation showing the interacting proteins with GCP6 in MCF10A cells with CRB3 knockdown. I. Representative images of immunofluorescent staining of GCP3 and GCP6 colocalization in MCF10A cells with the corresponding fluorescence intensity profile. GCP3 (red), GCP6 (green), and DNA (blue). (At least 3-4 random micrographs were analyzed for each condition in each of three independent experiments, scale bars, 10 μm) J. The comparison of cytoplasmic extracts from MCF10A cells and cells with CRB3 knockdown after fractionation in sucrose gradients. The γTuSC sedimentation was mainly in fractions 3, and γTuRC sedimentation was mainly in fractions 6. Bars represent the means ± SD; Unpaired Student's *t* test, ****P*<0.001.

and Rab11-positive endosomes was significantly decreased (**Fig. 5C, D**). We have previously reported that CRB3 regulates cell contact inhibition, which is significantly downregulated in confluent MCF10A cells[6]. Moreover, dynasore could rescue the CRB3 downregulation in confluent MCF10A cells (**Fig. 5E**). The knockdown of CRB3 did not affect Rab11 expression (Supplementary Fig. 4C). These results suggest that CRB3 is highly dynamic among the endosome system, and in particular, Rab11-positive endosomes are involved in the intracellular trafficking of CRB3.

CRB3 knockdown destabilizes γTuRC assembly during ciliogenesis

According to our findings, we assumed that CRB3 might be involved in polarized vesicle trafficking. The γ-Tubulin ring complex (γTuRC) is a central regulator of microtubule nucleation and a nucleation site of α-Tubulin and β-Tubulin at the microtubule[organizing center in mitotic cells[27]. Additionally, the primary cilium, a microtubule-based structure attached to the basal body, is transformed from the mother centriole. As we found that CRB3 knockdown disturbed the accumulation of γ-tubulin prompted us to analyze γTuRC assembly in ciliogenesis. The γTuRC is composed of multiple copies of the γ-tubulin small complex (γTuSC), GCP2 and GCP3, plus GCP4, GCP5, and GCP6[27, 28] (**Fig. 5F**). Our findings indicate that CRB3 downregulation did not affect the expression of γTuRC subunits (**Fig. 5G**). However, we performed co-IPs directed against GCP6 in CRB3-depleted cells and assessed the relative levels of both γTuSC-specific proteins compared with the control shRNA-treated cells. Interestingly, CRB3 knockdown significantly reduced the interaction of GCP6 with GCP3 (**Fig. 5H**). To verify this phenomenon, we observed the localization of GCP3 with GCP6 foci in MCF10A cells. Correspondingly, CRB3 downregulation significantly disturbed the colocalization of GCP3 with GCP6 foci in quiescence cells (**Fig. 5I**). To investigate whether CRB3 facilitate γTuRC assembly, we fractionated MCF10A cytoplasmic proteins on sucrose gradients according to published methods[29]. We noticed that γTuSC were sedimented mainly in fractions 3, while γTuRC sedimented in fractions 6 (**Fig. 5J**). However, CRB3 downregulation caused most of GCP6 and GCP5 to be precipitated independently of γTuRC in the low-density fractions (**Fig. 5J**). Together, these results demonstrate that the depletion of CRB3 disrupts molecular interactions between the γTuRC proteins, resulting in a failure of microtubule formation in the primary cilium. However, this depletion may not affect the expression of these molecules during γTuRC assembly. Based on these findings, we hypothesize that CRB3 could facilitate the polarized vesicle trafficking of some γTuRC-specific proteins through Rab11-positive endosomes.

CRB3 interacts with Rab11

To fully prove this hypothesis, we first examined the interaction of CRB3 with Rab11-positive endosomes and γTuRC-specific proteins in MCF10A cells. Using CRB3 as bait, we identified that CRB3 could interact with Rab11, but not GCP6 and GCP3 (**Fig. 6A**). In contrast, Rab11 could interact with CRB3 and GCP6 instead of GCP3 by tagging Rab11 as bait (**Fig. 6B**). The co-IP results showed that CRB3 knockdown did not affect Rab11 interacting with GCP6, but the level of Rab11 binding to GCP6 tended to decrease in MCF10A cells with CRB3 knockdown (**Fig. 6C**). Consistent with other reports[30], Rab11 knockdown significantly inhibited primary cilium formation in MCF10A cells (Supplementary Fig. 5A-C).

Based on these results, we further detected the region of CRB3b that interacts with Rab11 by using coexpression and co-IP in HEK293 cells. CRB3b is composed of a signal peptide (SP), extracellular domain, transmembrane (TM), FERM-binding domain (FBD), and carboxy-terminal PDZ-binding domain (PBD). We constructed CRB3-GFP fusion proteins by fusing GFP tags to the extracellular domain, as previously reported[31] (**Fig. 6D**). According to these domains, we also constructed the truncations of CRB3-GFP fusion proteins with serial C-terminal deletions (**Fig. 6E**) and another Flag-Rab11a fusion protein. The coexpression and co-IP results showed that

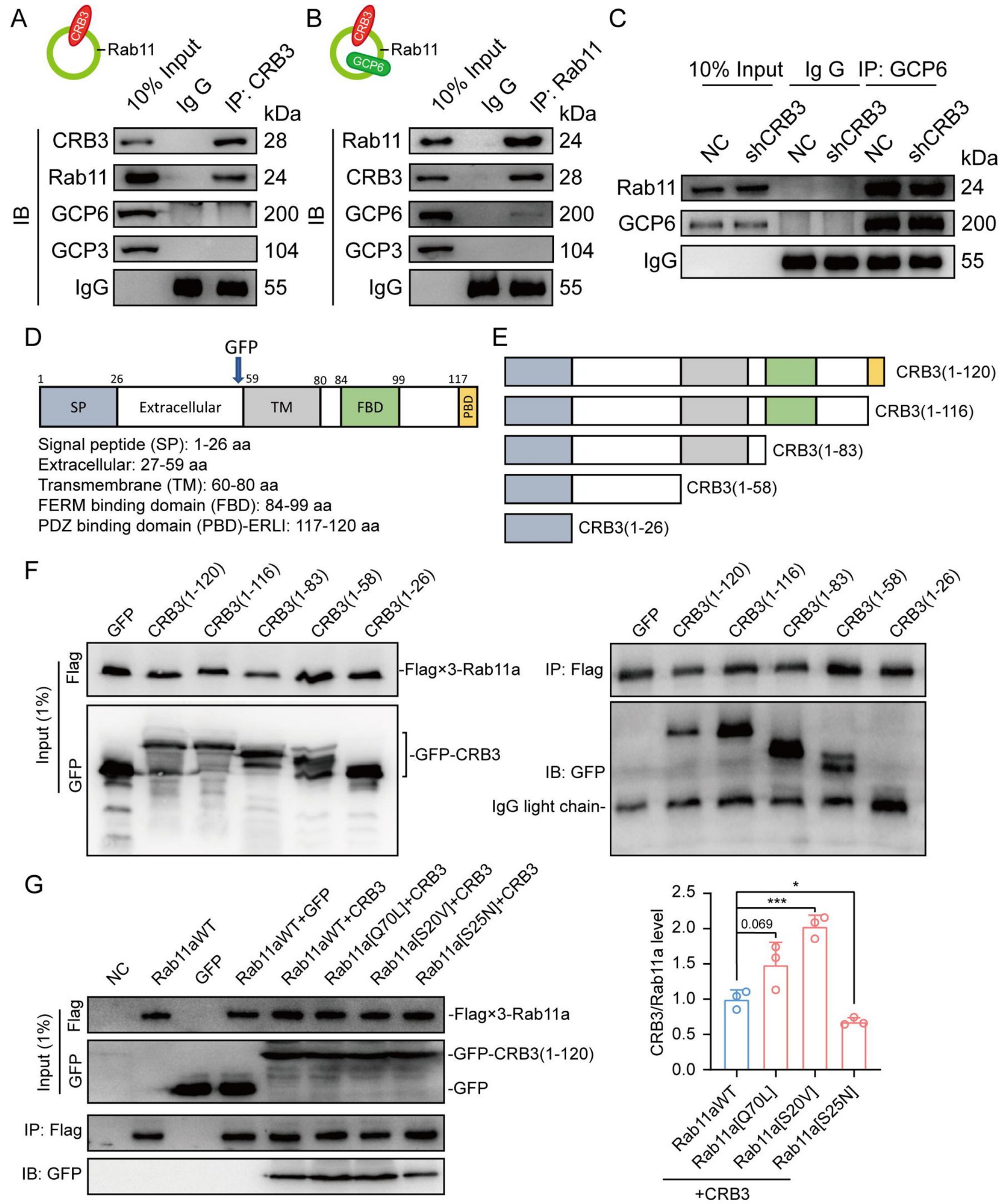

Figure 6.

CRB3 interacts with Rab11.

A. Coimmunoprecipitation of CRB3 with Rab11, GCP6 and GCP3 in MCF10A cells. B. Coimmunoprecipitation of Rab11 with CRB3, GCP6 and GCP3 in MCF10A cells. C. Coimmunoprecipitation of Rab11 with GCP6 in control and CRB3 knockdown MCF10A cells. D. Schematic diagram of CRB3b domains. E. Diagram truncations of CRB3b-GFP fusion proteins with serial C-terminal deletions. F. Domain mapping of CRB3b-GFP for Flag-Rab11a binding. Flag antibody co-IP of the full-length CRB3b-GFP and truncations of CRB3b-GFP with Flag-Rab11a were cotransfected into HEK293 cells for 48 h. Immunoblot analysis was performed using GFP and Flag antibodies. G. Coimmunoprecipitation of Rab11 mutant variants with full-length CRB3b-GFP. Flag antibody co-IP of the full-length CRB3b-GFP with Flag-Rab11aWT, Flag-Rab11a[Q70L], Flag-Rab11a[S20V] and Flag-Rab11a[S25N] were cotransfected into HEK293 cells for 48 h. Immunoblot analysis was performed using GFP and Flag antibodies. Bars represent the means ± SD, and the experiments were performed in triplicate; Unpaired Student's *t* test, **P*<0.05, ****P*<0.001.

Flag-Rab11a interacts with full-length CRB3 (1-120), CRB3 (1-116), CRB3 (1-83), and CRB3 (1-58) but not CRB3 (1-26) (**Fig. 6F**). Therefore, the region of CRB3b that interacts with Rab11 is amino acids 27 to 58.

Rab11 is a small GTP-binding protein that plays an essential role in regulating the dynamics of recycling endosomes, which must undergo GDP/GTP cycles. In particular, Rab11a[S25N] is a GDP-locked form, while Rab11a[S20V] and Rab11a[Q70L] are GTP-locked forms[30]. To further investigate the interaction between the GTP-bound form of Rab11a and CRB3b, we constructed these mutant variants of Rab11a. Next, we found that CRB3b interacts more strongly with Rab11a[Q70L] and Rab11a[S20V], while weakly interacting with Rab11a[S25N], compared to Rab11aWT (**Fig. 6G**). Together, these results indicate that the interaction of CRB3b and Rab11a is dependent on the GTPase activity of Rab11a.

CRB3 navigates GCP6/Rab11 trafficking vesicles to CEP290 in the primary cilium

Since CRB3 knockdown does not affect the interaction with Rab11-positive endosomes and γTuRC-specific proteins GCP6, our focus was to determine whether CRB3 affects the localization of GCP6/Rab11 trafficking vesicles to the basal body of the primary cilium. We reviewed the identification of CRB3 interacting proteins and found some centriolar proteins. CEP290, for example, is located in the transition zone of the primary cilium and is required for the formation of microtubule-membrane linkers[32]. Then, we verified that exogenous CRB3b could bind to CEP290, Rab11, GCP6 (**Fig. 7A**), and Rab11 also interacted with CEP290, CRB3-GFP, GCP6 (**Fig. 7B**). Similar to **Fig. 6C**, the amount of GCP6 bound to Rab11 was reduced within CRB3 knockdown. Importantly, CRB3 knockdown made it difficult for Rab11 to bind CEP290 (**Fig. 7C**). We corroborated these findings by checking the colocalization of the basal body foci of GCP6 and γ-tubulin, γ-tubulin and Rab11 in MCF10A and MEF cells. In quiescence control cells, GCP6 foci and γ-tubulin formed a prominent focus at the basal body of the primary cilium, as observed by immunofluorescence. However, CRB3 downregulation eliminated this focus and significantly disturbed the basal body foci of GCP6 and γ-tubulin in MCF10A cells (**Fig. 7D**), and γ-tubulin and Rab11 in MEF cells (**Fig. 7E**). At the beginning of this experiment, we also attempted to show the interaction of CRB3, Rab11 and GCP6 by expressing fluorescence tagged GCP6 protein and using FRET assay. However, the centrosomal foci of fluorescent GCP6 was difficult to capture due to weak signal, and this may require re-overexpression of ninein-like protein (Nlp) to promote γTuRC enrichment[29, 33]. Although CRB3 knockdown had little effect on the formation of GCP6/Rab11 trafficking vesicles, it significantly inhibited γTuRC assembly during ciliogenesis. This, in turn, impacted the transport of GCP6/Rab11 trafficking vesicles to the basal body of the primary cilium. Since CRB3 affected the binding of Rab11 and CEP290, we speculate that CRB3 may act as a navigator, navigating GCP6/Rab11 trafficking vesicles to the basal body of the primary cilium.

CRB3 regulates the Hh and Wnt signaling pathways in tumorigenesis

To investigate the role of CRB3 in regulating ciliary assembly in breast cancer, we examined the relationship between CRB3 and the primary cilium in breast cancer tissues. We found that CRB3 was localized at the subapical surface of the mammary gland lumen in the paracarcinoma tissues, but no obvious expression or localization in breast cancer tissues (**Fig. 8A**). Immunofluorescence analysis showed that the intact primary cilium was observed in the adjacent para-carcinoma tissues, whereas the number of cells with primary cilia was much lower in breast cancer tissues (**Fig. 8B, C**). Consistent with previous findings, CRB3 was localized at the basal body of the primary cilium in adjacent para-carcinoma tissues (**Fig. 8D**). These results again suggest that a defect in CRB3 expression inhibits ciliary assembly, which could be involved in tumorigenesis.

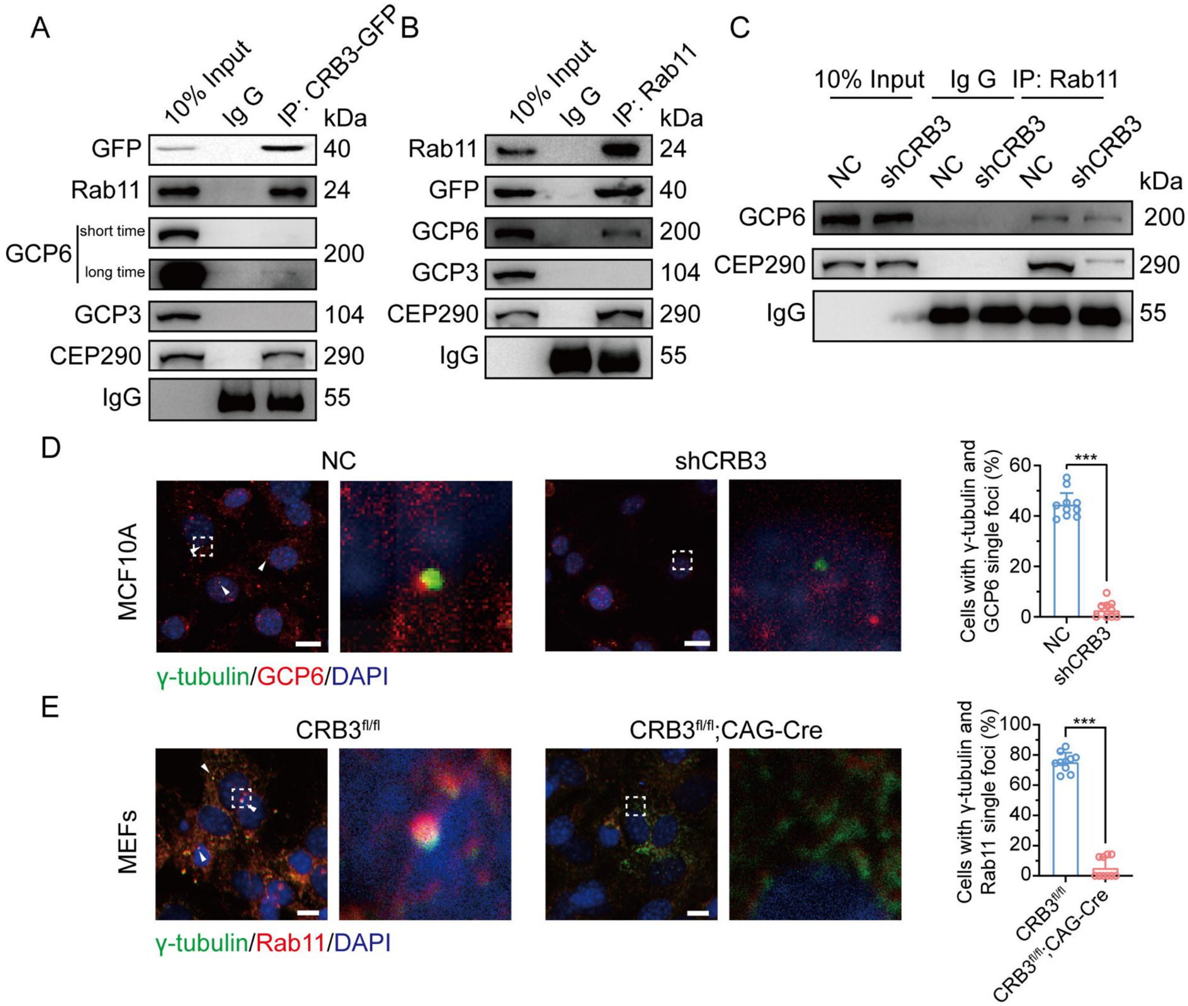

Figure 7.

CRB3 navigates GCP6/Rab11 trafficking vesicles to the basal body of the primary cilium.

A. Coimmunoprecipitation of exogenous CRB3b with Rab11, GCP6, GCP3 and CEP290 in MCF10A cells. B. Coimmunoprecipitation of Rab11 with exogenous CRB3b, GCP6, GCP3 and CEP290 in MCF10A cells. C. Coimmunoprecipitation of Rab11 with GCP6 and CEP290 in control and CRB3 knockdown MCF10A cells. D. Representative immunofluorescent images of GCP6 and γ-tubulin colocalization of the basal body foci in MCF10A cells with CRB3 knockdown. γ-tubulin (green), GCP6 (red), and DNA (blue). (Colocalization is marked by arrows; scale bars, 25 μm) E. Representative immunofluorescent images of Rab11 and γ-tubulin colocalization of the basal body foci in MEF cells from *Crb3*$^{fl/fl}$ and *Crb3*$^{fl/fl}$;CAG-*Cre* mice. γ-tubulin (green), Rab11 (red), and DNA (blue). Foci are marked by arrows; scale bars, 25 μm. Bars represent the means ± SD; Unpaired Student's *t* test, ****P*<0.001.

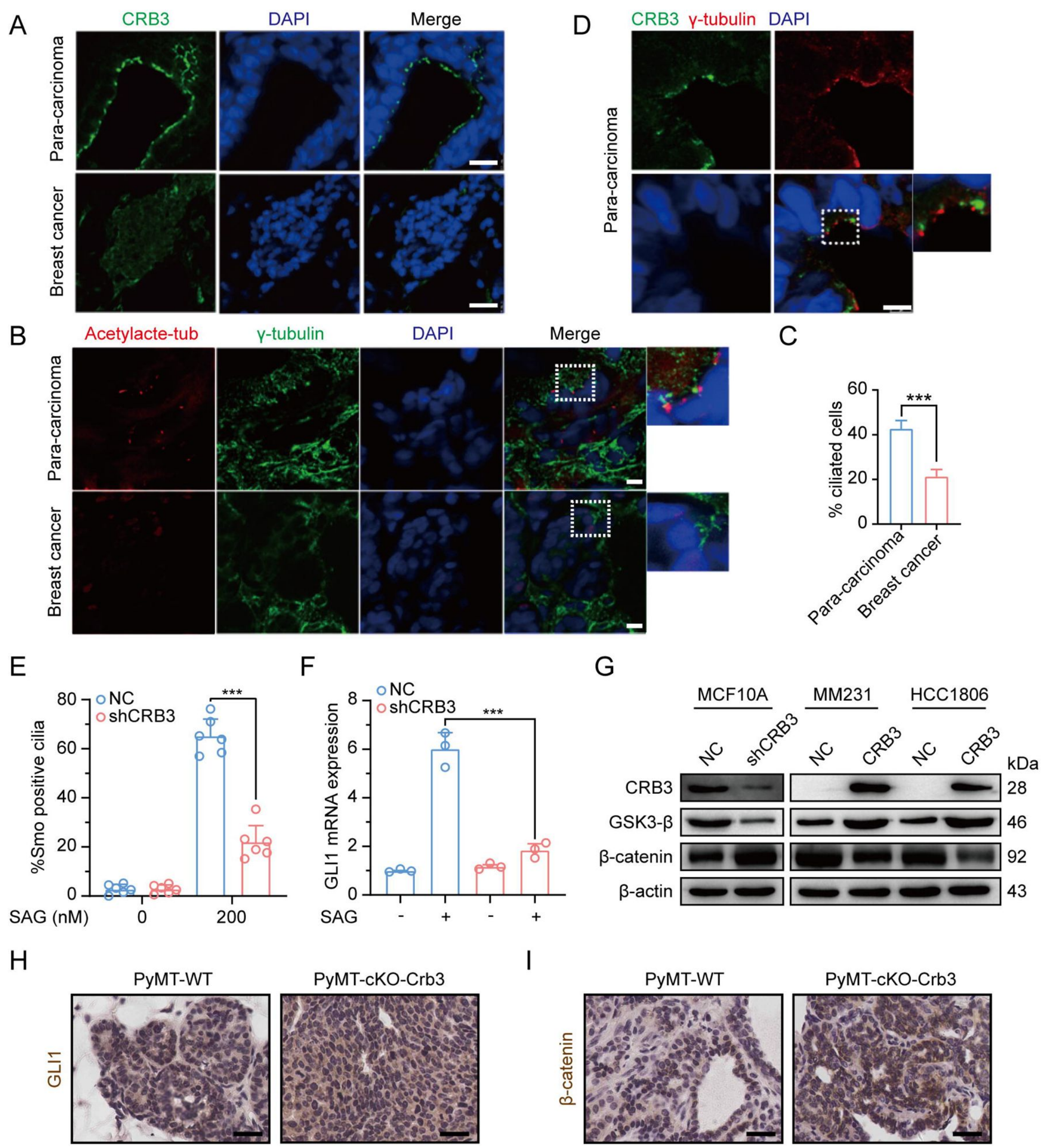

Figure 8.

Defects in CRB3 expression inhibit ciliary assembly in breast cancer tissues, and activate the Wnt signaling pathway in mammary cells and PyMT mouse model.

A. Representative immunofluorescent images of CRB3 in breast cancer tissues (n=50). CRB3 (green) and DNA (blue). (scale bars, 25 μm) B. Representative immunofluorescent images of the primary cilium in breast cancer tissues (n=50). Acetylated tubulin (red), γ-tubulin (green), and DNA (blue). (scale bars, 10 μm) C. Quantification of the proportion of cells with primary cilium formation in breast cancer tissues (n=50). D. Representative immunofluorescent images of CRB3 and γ-tubulin colocalization in adjacent paracarcinoma tissues. CRB3 (green), γ-tubulin (red), and DNA (blue). (scale bars, 25 μm) E. Quantification of the proportion of MFC10A cells with SMO translocation after CRB3 knockdown (n=6). F. Real-time quantitative PCR showing the relative mRNA expression of GLI1 upon SAG treatment in CRB3-depleted MFC10A cells (n=6). G. Immunoblot analyses of the effect of CRB3 on the molecules of the Wnt signaling pathway in mammary cells, and the experiments were performed in triplicate. H. I. Immunohistochemical analyses of GLI1 and β-catenin in primary tumors from PyMT-WT and PyMT-cKO-*Crb3* mice at 9 weeks old, respectively. (scale bars, 25 μm) Bars represent the means ± SD; Unpaired Student's *t* test, ****P*<0.001.

To investigate the effects of primary cilium absence, we examined alterations in the Hedgehog (Hh) and Wnt signaling pathways in CRB3-depleted cells. Firstly, we detected the translocation of SMO into the primary cilium and the expression of the target gene GLI1. SAG, a potent Smoothened (SMO) receptor agonist, activates the Hh signaling pathway. However, in MCF10A cells treated with SAG, CRB3 knockdown did not promote SMO translocation into the primary cilium (**Fig 8E**). After SAG treatment, the mRNA expression of GLI1 was significantly decreased in CRB3-depleted cells (**Fig 8F**). Crb3 knockout did not alter GLI1 expression and subcellular localization in primary tumors of PyMT-cKO-Crb3 mice (**Fig 8H**). We previously reported that CRB3 expression was high in immortalized mammary epithelial cells, whereas loss of CRB3 expression was observed in breast cancer cells[6]. On the other hand, western blotting showed that CRB3 knockdown resulted in significant downregulation of GSK3-β in MCF10A cells. Additionally, β-catenin, the major effector of the canonical Wnt signaling pathway, was upregulated. In breast cancer cells, CRB3b overexpression led to upregulation of GSK3-β and downregulation of β-catenin (**Fig. 8G**, Supplementary Fig. 6). Notably, β-catenin was significantly upregulated and markedly localized in the nucleus of PyMT-cKO-Crb3 mice (**Fig 8I**). Overall, these data suggest that CRB3 knockdown cannot activate the Hh signaling pathway but can activate the Wnt signaling pathway, which is ultimately contributes to tumorigenesis.

Discussion

Apical-basal cell polarity is essential for cellular architecture, development and homeostasis in epithelial tissues. However, most malignant cancers lack cell polarity due to unconstrained cell cycle and cell migration with decreased adhesion. CRB3, an important apical protein, is significantly reduced in various tumor cells and tissues, and promotes metastasis and tumor formation in nude mice[5, 6, 8, 34–36]. However, it has also been reported that CRB3 sequencing results are upregulated in breast cancer samples[37]. Here, we generated a novel transgenic mouse model with conditional deletion of Crb3 to investigate the role of polarity proteins in tumorigenesis. *Crb3* knockout mice were reported to have cystic kidneys, improper airway clearance in the lung, villus fusion, apical membrane blebs, and disrupted microvilli in intestine airway clearances, which demonstrated that CRB3 is crucial for the development of the apical membrane and epithelial morphogenesis[19, 38]. Importantly, our studies in the *Crb3* knockout mouse model suggest that, in addition to death after birth, it is characterized by smaller size, ocular abnormality, bronchial smooth muscle layer defect, reduced pancreatic islets and hyperplasia mammary ductal and renal tubular epithelium (some data not shown). These important phenotypic changes are due to disruption of epithelial polarity homeostasis, which may ultimately lead to tumorigenesis.

In addition, we observed the absence of primary cilia in the mammary ductal lumen and renal tubule in the *Crb3* knockout mouse model. Consistent with other reports, CRB3 knockdown regulated ciliary assembly in MDCK cells[17, 18]. Furthermore, CRB3 was localized in the inner segments of photoreceptor cells and concentrated with the connecting cilium throughout the development of the mouse retina[39]. These ocular defect phenotypes closely resemble those of EHD1 knockout mice, which regulate ciliary vesicle formation in primary cilium assembly[25, 40, 41]. Hence, CRB3 can affect primary cilium formation in various epithelial tissues, such as the breast, kidney, and retina.

Previous literature has reported that CRB3 localizes in the primary cilium itself in differentiated MDCK cells[17, 18], but we mainly found that it is not located on the basal body of the primary cilium in mammary epithelial cells. The immunofluorescence results in these literatures showed that CRB3 was scattered on the primary cilia, with a strong foci at the basal body. Notably, in rat kidney collecting ducts, the localization of CRB3 on primary cilia was significantly reduced, while the obvious localization was at the basal body[18]. Another study also reported the co-localization of CRB3b and γ-tubulin in MDCK cells[17]. This finding is consistent with our conclusion, and we also verified its co-localization with the centrosome by overexpressing CRB3b in mammary epithelial cells, indicating that CRB3 mainly localizes to the basal body of the primary cilium. Since CRB3b also alters primary cilium formation in the mammary epithelium, it is important to investigate the molecular mechanisms by which CRB3 regulates primary cilium formation.

Polarized vesicular traffic plays a crucial role in extending the primary cilium on apical epithelial surfaces. However, the molecular mechanism and its relationship to tumorigenesis remain unclear. Here, we reveal that Rab11-positive endosomes mediate the intracellular trafficking of CRB3 and that CRB3 can navigate GCP6/Rab11 trafficking vesicles to CEP290, promoting correct γTuRC assembly during primary cilium formation. When cells form tight junctions, CRB3 is mainly localized to the cell membrane, and subsequently the endosomes participate in the intracellular degradation process of CRB3 on the cell membrane. The intracellular CRB3 can bind to Rab11 through the extracellular domain, which in turn participates in primary cilium assembly. Additionally, in the presence of CRB3, the epithelium can form contact inhibition and an intact primary cilium in quiescence to achieve cellular homeostasis. In contrast, the loss of contact inhibition and primary cilium formation with CRB3 deletion can activate the Wnt signaling pathway, affect the Hh signaling pathway, and disrupt the imbalance of this cellular homeostasis, resulting in significant cell proliferation during tumorigenesis (**Fig. 9**).

The formation of the primary cilium has been divided into several distinct phases. It begins with the maturation of the mother centriole, also known as the centriole-to-basal-body transition[9, 42]. DAVs, which are small cytoplasmic vesicles originating from the Golgi and the recycling endosome, accumulate in the vicinity of the distal appendages of the mother centriole[25, 43, 44]. In growing RPE1 cells, Rab8a is localized to cytoplasmic vesicles and the Golgi/trans-Golgi network, while Rab8-positive vesicles are rapidly redistributed to the mother centriole by binding to Rab11-positive recycling and endosomes under serum starvation[26, 45]. The EHD1 protein converts these DAVs to form larger ciliary vesicles, which elongate through continuous fusion with Rab8-positive vesicles to produce the primary cilium membrane[25, 30, 41]. Therefore, Rab11-positive recycling endosomes, as important transporters, are necessary for primary cilium formation and centriole-basal-body transition in the early phase. Our study found that CRB3 could bind to some members of Rab small GTPase family, significantly participating in Golgi vesicle transport and vesicle organization pathways through mass spectrometry identification. And coexpression and co-IP results showed that CRB3 interacted with Rab11 via the extracellular domain, which mediated the intracellular trafficking of CRB3 within Rab11-positive endosomes at different cell densities.

Unlike motile cilia, primary cilia have a 9+0 structure, consisting of only nine outer doublet microtubules without a central pair. Similar to microtubule assembly at microtubule organizing centers (MTOCs) during mitosis, γ-tubulin, which is a part of γTuRC, is also concentrated in the basal body of the cilium in interphase[46]. γ-tubulin complexes have been shown to form microtubule templates and regulate microtubule nucleation through longitudinal contacts with α[tubulin and β[tubulin[27]. The γTuRC consists of multiple copies of γTuSC, which is composed of two copies of γ-tubulin and one each of GCP2 and GCP3. Then, GCP2 and GCP3 interact with GCP4, GCP5 and GCP6 to form γTuRC[47]. Our experiments showed that CRB3 knockdown did not affect the expression of the γTuRC-specific proteins GCP3, GCP6 and γ-tubulin, but inhibited the interaction between GCP6 and GCP3, leading to destabilization of γTuRC. Further analysis revealed that significant colocalization was observed between GCP6- and Rab11-positive vesicles at low cell density, while at high cell density, GCP6 accumulated at the centrosome with GCP3. CRB3 knockdown resulted in the disappearance of this accumulated GCP6. These results indicated that GCP6/Rab11 trafficking vesicles must be transported to the basal body for ciliogenesis. While it is still unclear how CRB3 interferes with γTuRC assembly during primary cilium formation, we

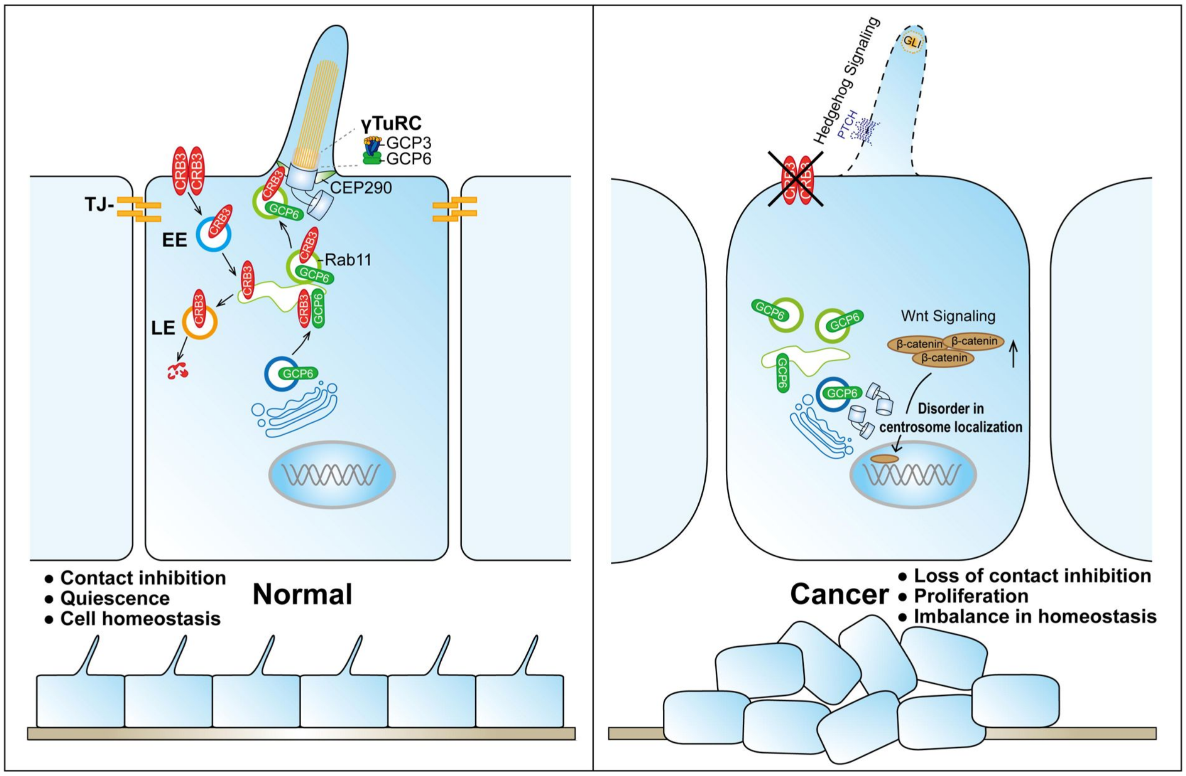

Figure 9.

Schematic model of CRB3 regulating ciliary assembly.

Graphic summary of prominent phenotypes observed after CRB3 deletion. CRB3 is localized on apical epithelial surfaces and participates in tight junction formation to maintain contact inhibition and cell homeostasis in quiescence. Inside the cell, Rab11-positive endosomes mediate the intracellular trafficking of CRB3, and CRB3navigates GCP6/Rab11 trafficking vesicles to CEP290. Then, GCP6 is involved in normal γTuRC assembly in ciliogenesis. In CRB3 deletion cells, the primary cilium fails to assemble properly, and the Wnt signaling pathway is activated through β-catenin upregulation and nuclear localization, but the Hh signaling pathway fails to be activated. This cellular imbalance is disrupted, leading to tumorigenesis. Abbreviations: γTuRC, γ-tubulin ring complex; TJ, tight junction; EE, early endosome; LE, late endosome.

hypothesize that Rab11-positive endosomes mediate the intracellular trafficking of CRB3 and that CRB3, as a navigator interacting with CEP290, can navigate GCP6/Rab11 trafficking vesicles to the transition zone for γTuRC assembly during primary cilium formation.

CEP290 is a centriolar or ciliary protein that localizes to the centrosomes in dividing cells, the distal mother centriole in quiescent cells, and the transition zone in the primary cilium[48]. Mutations in *CEP290* are associated with various diseases, such as the devastating blinding disease Leber's Congenital Amaurosis (LCA), nephronophthisis, Senior Løken syndrome (SLS), Joubert syndrome (JS), Bardet-Biedl syndrome (BBS), and lethal Meckel-Gruber syndrome (MGS)[49]. In particular, the phenotypes of CEP290-associated ciliopathies are very similar to those in our *Crb3* knockout mouse model, with lethality and ocular and renal abnormalities. CEP290 knockdown significantly decreased the number of cells with primary cilia, and CEP290 knockout mice showed mislocalization of ciliary proteins in the retinas[50, 51]. Additionally, depletion of CEP290 disrupts protein trafficking to centrosomes and affects the recruitment of the BBSome and vesicular trafficking in ciliary assembly[52, 53]. Although CEP290 serves as a hub to recruit many ciliary proteins, it has not been reported to be involved in polarized vesicle trafficking or localization. Our study verified that CEP290 could interact with the polarity protein CRB3. CRB3 knockdown disrupts the interaction between CEP290 and Rab11-positive endosomes. Thus, these results validate our hypothesis. Although CRB3 navigates GCP6/Rab11 trafficking vesicles to CEP290, we further need to detect the region of CRB3 interacting with CEP290 and what other cargos are transported by CRB3/Rab11 trafficking vesicles in ciliary assembly.

In addition, our results showed that CRB3 knockdown failed to activate the Hh signaling pathway with SAG in MCF10A cells. Under untreated conditions, PTCH1 located at the ciliary membrane inhibits SMO, and Suppressor of Fused (SuFu) sequesters Gli transcription factors, leading to their degradation. SAG can activate the Hh signaling pathway, resulting in dissociation into the nucleus to regulate the expression of downstream target genes[54]. The Hh signaling pathway mainly regulates cellular growth and differentiation and is abnormal in different types of tumors. Some studies indicate that cilia are double-sided, promoting and preventing cancer development through the Hh signaling pathway *in vivo*. It is thought that cilia deletion could activate SMO to inhibit tumor growth, while promoting carcinogenesis was induced by activated Gli2[55]. Although SAG cannot activate the Hh signaling pathway in CRB3-depleted cells, further research is needed to determine the role of CRB3 in the carcinogenic process through the ciliary Hh signaling pathway. Additionally, another canonical Wnt signaling pathway was altered after CRB3 knockdown. The Wnt signaling pathway is an important cascade regulating development and stemness, and aberrant Wnt signaling has been reported in many cancer entities, especially colorectal cancer[56]. In our study, CRB3 downregulated the expression of β-catenin. Thus, CRB3 can affect cilium-related signaling pathways in tumorigenesis.

In summary, our study provides novel evidence for the role of the apical polarity protein CRB3 in regulating viability, mammary and ocular development, and primary cilium formation in germline and conditional knockout mice. Deletion of CRB3 caused irregular lumen formation and ductal epithelial hyperplasia in mammary epithelial-specific knockout mice. Further study uncovered a new function of CRB3b in interacting with Rab11, navigating GCP6/Rab11 trafficking vesicles to CEP290 for γTuRC assembly during ciliogenesis. In addition, CRB3 also participates in regulating cilium-related Hh and Wnt signaling pathways in tumorigenesis. Understanding these polarity protein-mediated vesicle trafficking mechanisms will shed new light on luminal development and tumorigenesis.

Methods

Generation of floxed *Crb3* mice

All animal experiments were verified and approved by the Committee of Institutional Animal Care and Use of Xi'an Jiaotong University. Heterozygous *Crb3*$^{wt/fl}$ mice (C57BL/6J) were generated using Cas9/CRISPR-mediated genome editing (Cyagen Biosciences, Guangzhou, China). The gRNA to *Crb3* and Cas9 mRNA was coinjected into fertilized mouse eggs to generate targeted conditional knockout offspring. The *Crb3* gRNA sequences were gRNA1, 5'-GGCTGGGTCCACACCTACGGAGG-3'; gRNA2, 5'-ACCCACAAAGCCACGCCA GTGGG-3'. Mouse genotyping was screened using PCR with the primers F1, 5'-ACATAAGGCCTTCCGTTAAGCTG-3'; R1, 5'-GTGGATTCGGACCAGTCTGA-3'. *Crb3*$^{wt/fl}$ mice were intercrossed with MMTV-*Cre* mice or CAG-*Cre* mice to generate tissue-specific *Crb3* knockout mice. MMTV-*Cre* mice (FVB), CAG-*Cre* mice (C57BL/6N), and MMTV-PyMT mice (FVB) were purchased from Cyagen Biosciences. All mice were bred in the specific pathogen-free (SPF) animal houses of the Laboratory Animal Center of Xi'an Jiaotong University.

Mice were genotyped using PCR and DNA gel electrophoresis. Genomic DNA from the mouse tail was extracted using the Mouse Direct PCR Kit (Bimake, TX, USA, #B40015) according to the manufacturer's instructions. The genotyping primers were F3, 5'-TTGAGAGTCTTAAGCAGTCAGGG-3'; R5, 5'-AACCTTTCCCAGGAGTA TGTGAC-3'. PCR results showed that *Crb3*$^{wt/wt}$ was one band with 163 bp, *Crb3*$^{wt/fl}$ was two bands with 228 bp and 163 bp, and *Crb3*$^{fl/fl}$ was one band with 228 bp. The primers for identifying *Cre* alleles were F, 5'-TTGAGAGTCTTAAGCAGTCAGGG-3'; R, 5'-TTACCACTCCCAGCAAGACAC-3', and amplification with one 247 bp band. The reaction conditions of *Crb3* and *Cre* PCR were as follows: 94°C for 5 min, 94°C for 20 s, 60°C for 30 s, and 72°C for 20 s for 35 cycles.

Mouse mammary gland whole mount analysis

After euthanizing the mice, the inguinal mammary gland tissues were removed intact and fully spread onto slides. The tissues were rapidly fixed in Carnoy's fixative for 2 h at room temperature. Subsequently, the tissues were washed in 70% ethanol solution for 15 min at room temperature, and in 50% and 30% ethanol and distilled water for 5 min each. Staining was performed in carmine alum solution at 4°C overnight. After washing with a 70% ethanol solution, the tissues were washed again with 70%, 95%, and 100% ethanol for 15 min each. Tissues were cleared in xylene overnight at room temperature and then mounted with neutral balsam. Images were obtained using a microscope (Leica DMi8, Wetzlar, Germany).

Cell culture and transfection

The MCF10A, MCF7, T47D, MDA-MB-231, HCC1806, T47D, and MDA-MB-453 cell lines were purchased from Shanghai Institute of Biochemistry and Cell Biology (Chinese Academy of Sciences, Shanghai, China). MCF10A cells were routinely grown in DMEM/F12 (1:1) media (HyClone, UT, USA) supplemented with 5% horse serum, 20 ng/ml human EGF, 10 μg/ml insulin, and 0.5 μg/ml hydrocortisone. MCF7, MDA-MB-231 and MDA-MB-453 cells were cultured in high glucose DMEM media (HyClone, UT, USA) supplemented with 10% FBS (HyClone, UT, USA, #SH30084.03). T47D and HCC1806 cells were grown in RPMI-1640 media (HyClone, UT, USA) supplemented with 10% FBS. All cell lines were incubated in 5% CO_2 at 37°C.

siRNAs were purchased from GenePharma company (Shanghai, China). Cultured cells were transfected with siRNAs or plasmids using Lipofectamine 2000 (Invitrogen, CA, USA, #11668-019) following the manufacturer's protocol. The negative control shRNA (NC) and shRNA against CRB3 (both isoforms CRB3a and CRB3b) were packaged into lentivirus as previously described[5, 6].

DNA constructs and stable cell lines

CRB3b was PCR amplified from RNA extracted from MCF10A cells and cloned into the pCMV-Blank vector (Beyotime, Shanghai, China, D2602) using T4 ligase (NEB, MA, USA). Then, the GFP sequence was inserted between the extracellular and transmembrane CRB3 domains to generate the pCMV-CRB3-GFP vector, as described previously[31]. The full-length CRB3 (1-120), CRB3 (1-116), CRB3 (1-83), CRB3 (1-58), and CRB3 (1-26) were amplified by PCR from the pCMV-CRB3-GFP vector using specific primers to create EcoR [and Xho [restriction sites, and then cloned into the pCMV-Blank vector. The full-length was also cloned into the lentiviral vector pLVX-TetOne-Puro (Clontech, Takara, #631849). The plasmid pECMV-3×FLAG-RAB11A was purchased from SinoBiological (Beijing, China). The plasmids of pECMV-3×FLAG-RAB11A[S20V]/[S25N]/[Q70L] mutants were constructed using the *Fast* Mutagenesis System (Transgen, Beijing, China, FM111-01).

Lentiviruses were produced in HEK293T cells. CRB3-GFP in MCF10A and MCF7 cells was generated using a lentivirus expression system. Stable cell lines were screened using puromycin after lentiviral infection for 72 h.

3D morphogenesis

To initiate 3D morphogenesis, growth factor-reduced Matrigel (Corning, USA, #354230) was added to a four-well chamber slide system (Corning, USA, #177437) and then incubated at 37°C for solidification. MCF10A cells were plated into this chamber slide and cultured in DMEM/F12 (1:1) media (HyClone, UT, USA) supplemented with 2% horse serum, 5 ng/ml human EGF, 10 μg/ml insulin, 0.5 μg/ml hydrocortisone, and 2.5% Matrigel. 3D morphogenesis was photographed using a microscope (Leica DMi8, Wetzlar, Germany) at different time points. On day 14 after cell culture, 3D MCF10A cells were fixed and stained.

Immunohistochemistry

The paraffin-embedded sections of samples were baked, deparaffinized in xylene, and sequentially rehydrated in gradient concentrations of ethanol, sequentially. Then, antigen retrieval was performed in Tris-EDTA buffer (pH 9.0) heated to 95°C for 20 min. Endogenous peroxidase activity was reduced in 3% hydrogen peroxide for 10 min at room temperature. Then, 5% goat plasma was used for blocking at 37°C for 30 min, and the sections were incubated with specific primary antibodies overnight at 4°C. Primary antibodies used were Ki67 (Abcam, UK, ab279653), phospho-histone H3 (Abcam, UK, ab267372), cleaved caspase 3 (CST, USA, #9661), GLI1 (CST, USA, #3538), and β-catenin (CST, USA, #8480). The following steps used biotin-streptavidin HRP detection kits (ZSGB-BIO, China, SP-9001, SP-9002) according to the manufacturer's instructions. The sections were stained with DAB (ZSGB-BIO, China, ZLI-9017), counterstained with hematoxylin, dehydrated in gradient concentrations of ethanol, cleared in xylene, and mounted with neutral balsam, sequentially. Images were obtained using a slide scanner (Leica SCN400, Wetzlar, Germany).

Immunofluorescence

Cells were plated on coverslips and fixed in 4% paraformaldehyde for 30 min. Then, the membranes were permeabilized using 0.2% Triton X-100. The cells were blocked with 5% BSA solution for 1 h at room temperature. Cells were incubated with specific primary antibodies overnight at 4°C. Caspase 3 (CST, USA, #9662S), α-tubulin (CST, USA, #3873), acetylated tubulin (Proteintech, USA, #662001-IG), γ-tubulin (Proteintech, USA, #15176-1-AP), pericentrin (Abcam, UK, ab4448), CRB3 (Sigma, USA, HPA013835), EEA1 (BD, USA, #610456), Rab11 (BD, USA, #610823), and GCP6 (Abcam, UK, ab95172) were used as primary antibodies. The secondary antibodies were Alexa Fluor 488-labeled or 594-labeled (Invitrogen, USA, A32731, A32744). DPIA (5 μg/ml) was used for DNA staining, and images were captured using a confocal microscope (Leica SP5 [, Wetzlar, Germany).

MEFs isolation and maintenance

The embryos were harvested from *Crb3*$^{wt/fl}$;CAG-*Cre* mice crosses at E13.5, and the head, tail, and internal organs were removed. The tail was used for genotyping, and DNA was extracted from it. Then, the body was minced and digested with trypsin in a 37°C incubator for 30 min. The resulting cells were resuspended in MEF media (DME, 10% FBS, 1% penicillin/streptomycin) and plated onto 10 cm cell culture dishes with one embryo in each. They were subsequently passaged 1:3 with MEF media to expand the culture.

Real-time PCR assay

Total RNA from cell lines was isolated using TRIzol reagent (Invitrogen, USA, #15596026), and 5 μg RNA was converted to cDNA with the RevertAid first strand cDNA synthesis kit (Thermo, USA, K1622) according to the manufacturer's instructions. Real-time PCR was prepared with SYBR qPCR Premix (Takara, Japan, RR420L), and then performed with a real-time PCR detection instrument (Bio–Rad CFX96, USA). The primers used for real-time PCR were purchased from Tsingke Biotech (Beijing, China), and the sequences are listed in Table 1. The mRNA expression was normalized to *GAPDH*, and fold changes were calculated using the ΔΔCt method.

Immunoprecipitation and immunoblotting

Cultured cells were lysed in RIPA buffer supplemented with protease inhibitors (Roche, NJ, USA). Then, the cell lysates were subjected to SDS–PAGE separation and were transferred to PVDF membranes. The membranes were subjected to immunoblot assays using antibodies against CRB3 (Santa Cruz, USA, sc-292449, both isoforms CRB3a and CRB3b can be detected), GCP2 (Santa Cruz, USA, sc-377117), GCP3 (Santa Cruz, USA, sc-373758), GCP4 (Santa Cruz, USA, sc-271876), GCP5 (Santa Cruz, USA, sc-365837), GCP6 (Abcam, UK, ab95172), γ-tubulin (Proteintech, USA, #66320-1-Ig), Rab11 (Abcam, UK, ab128913), GFP (Roche, USA, #11814460001), Flag (Sigma, USA, F7425), CEP290 (Santa Cruz, USA, sc-390637), GSK3-β (CST, USA, #12456), β-catenin (CST, USA, #8480), GAPDH (Proteintech, USA, #HRP-6004), and β-actin (Proteintech, USA, #HRP-60008). HRP-conjugated secondary antibodies (CST, USA, #7074, #7076) were then incubated at room temperature for 1 h in the dark. Final detection was achieved by using ECL Plus (Millipore, Germany, WBULS0500). Proteins for the immunoprecipitation assay were lysed with NP40 buffer. The cell lysates were detected by a Dynabeads protein G immunoprecipitation kit (Invitrogen, USA, #10007D) following the manufacturer's instructions. The elution protein complexes were subjected to an immunoblot assay.

Protein identification and bioinformatics analyses

LC-MS/MS analysis was conducted by PTM Bio (Zhejiang, China). Q Exactive^{TM} Plus (Thermo, MA, USA) was used for tandem mass spectrometry data analysis. The Blast2GO program against the UniProt database was used to perform functional annotation and classification of the identified proteins. Subsequently, DAVID tools were used to perform pathway enrichment analysis.

Scanning electron microscope (SEM) observations

MCF10A cells with CRB3-GFP and CRB3 knockdown were plated on culture slides, and doxycycline (Dox) was added to induce the expression of CRB3-GFP. After the cells were fully confluent, the slides were fixed with 4% paraformaldehyde at 4°C overnight, followed by 1% osmium tetroxide at 4°C for 1 h. Dehydration was performed at room temperature, and the samples were dried using the critical point drying method. Images were obtained by using a scanning electron microscope (Hitachi TM1000, Japan).

Clinical data

Patient data, breast cancer tissues, and adjacent para-cancerous tissues were collected from the First Affiliated Hospital of Xi'an Jiaotong University (Shaanxi, China). All patients signed informed consent forms before surgery. This research was authorized by the Ethics Committee of the First Affiliated Hospital of Xi'an Jiaotong University and conducted in conformity with the Declaration of Helsinki.

Statistics

Statistical analysis was performed using SPSS statistics 23.0 for Windows (IBM, Armonk, USA). All experiments were repeated at least three times. The values are expressed as the mean ± standard deviation (SD). Unpaired Student's *t* test was used to compare the differences between two groups. One-way ANOVA followed by Dunnett's multiple comparisons test was used for multiple comparisons. The χ^2 test was used to assess the significance of the observed frequencies. $P<0.05$ was considered an indicator of statistical significance.

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Conflict of interest

The authors declare no conflicts of interest.

Author contributions

P.L. designed the research; B.W., Z.L., T.T, Y.J., Y.S., R.W., H.C., J.L. and P.L. performed the experimental work, and M.Z., X.G., J.L. and S.S. analyzed the data; B.W., Z.L., Y.R. and P.L. wrote the original manuscript. P.L. conceived the study and critically revised the manuscript. All authors read and approved the final manuscript.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Supplementary Figure 1. Generation of *Crb3* knockout mice using the Cre-loxP system.

A. Schematic representation of the strategy for loxP sequence insertion and PCR genotyping. The loxP sites (gray arrow) flanked either side of exon 3 (orange) in the Crb3 gene. The F3&R5 and F4&R6 primer pair were used to differentiate the alleles. B. Southern blot analysis of a positive ES cell clone confirmed homologous recombination. The genomic DNA from ES cell clones and wild-type cells was digested with Avr or Sac and hybridized with a 5' external probe or a 3' internal probe. C. PCR identification of mouse genotyping showing *Crb3*$^{wt/wt}$, *Crb3*$^{wt/fl}$ and *Crb3*$^{fl/fl}$ alleles. D. The expression of Crb3 was detected by immunoblotting in mammary gland tissues in *Crb3*$^{fl/fl}$ and *Crb3*$^{fl/fl}$;MMTV-*Cre* mice at 8 weeks old. E. Real-time quantitative PCR showing the relative mRNA expression of *Crb3* in mammary gland tissues (n=10 mice). F. Crb3 expression was assessed using immunohistochemistry in mammary gland tissues. The mammary epithelial cells were marked by areas of white dotted lines. (scale bars, 50 μm) Bars represent the means ± SD; Unpaired Student's *t* test, ***P<0.001.

Supplementary Figure 2. CRB3 knockdown promotes proliferation of mammary epithelial cells, while overexpression promotes apoptosis of breast cancer.

A. B. Immunoblot and real-time quantitative PCR analysis of CRB3 knockout efficiency in MCF10A cells. C. MCF10A proliferation assay over six successive days. D. Cell cycle showing the distribution of MCF10A cells. E. Quantification of cell cycle distribution. F.G.H. Apoptosis of MCF7, T47D, and MDA-MB-453 breast cancer cells with CRB3b overexpression and quantification of apoptotic cells. The data are three independent experiments performed in triplicate. Bars represent the means ± SD; Unpaired Student's *t* test, * P<0.05, **P<0.01.

Supplementary Figure 3. CRB3 alters primary cilium formation in MEFs from *Crb3*$^{fl/fl}$;CAG-*Cre* mice.

A. Immunoblot analysis of CRB3b conditional overexpression upon dox induction efficiency in MCF7 cells. B. Representative immunofluorescent staining of primary cilium formation in MEFs from *Crb3*$^{fl/fl}$ and *Crb3*$^{fl/fl}$;CAG-*Cre* mice. Acetylated tubulin (red), γ-tubulin (green), and DNA (blue). (scale bars, 10 μm) C. Quantification of the proportion of cells with primary cilium formation (At least 3-4 random micrographs were analyzed for each condition in each of three independent experiments, n=10). Bars represent the means ± SD; Unpaired Student's *t* test, ***P<0.001.

Supplementary Figure 4. The effect of CRB3 on the expression of some ciliogenesis-related genes and Rab11.

A. Quantification of mRNA expression of ciliogenesis-related genes in MCF10A cells (normalized to GAPDH, data are three independent experiments performed in triplicate). B. Coomassie blue staining showing the gel electrophoresis of exogenous CRB3b immunoprecipitated precipitates. C. Immunoblot analysis of the effect of CRB3 on Rab11 in MCF10A cells. Bars represent the means ± SD; Unpaired Student's *t* test.

Supplementary Figure 5. Rab11 knockdown caused defects in primary cilium.

A. Representative images of immunofluorescent staining of primary cilium formation with Rab11 knockdown in MCF10A cells. Acetylated tubulin (red), γ-tubulin (green), and DNA (blue). (primary cilium is marked by arrows; scale bars, 20 μm) B. Quantification of the proportion of cells with primary cilium formation in MCF10A cells (At least 3-4 random micrographs were analyzed for each condition in each of three independent experiments, n=10). C. Immunoblot analysis of Rab11 knockout efficiency in MCF10A cells. Bars represent the means ± SD; Unpaired Student's *t* test, ***P<0.001.

Supplementary Figure 6. Western blotting quantification of the molecules in the Wnt signaling pathway.

A. The relative protein levels of CRB3, GSK3-β, and β-catenin in CRB3-depleted MFC10A cells (normalized to β-actin). B. The relative protein levels of CRB3, GSK3-β, and β-catenin in MDA-MB-231 and HCC 1806 breast cancer cells with CRB3b overexpression (normalized to β-actin). The data are three independent experiments performed in triplicate. Bars represent the means ± SD; Unpaired Student's *t* test, **P<0.01, ***P<0.001.

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Reviewer #1 (Public Review):

In this study the authors first perform global knockout of the gene coding for the polarity protein Crumbs 3 (CRB3) in the mouse and show that this leads to perinatal lethality and anopthalmia. Next, they create a conditional knockout mouse specifically lacking CRB3 in mammary gland epithelial cells and show that this leads to ductal epithelial hyperplasia, impaired branching morphogenesis and tumorigenesis. To study the mechanism by which CRB3 affects mammary epithelial development and morphogenesis the authors turn to MCF10A cells and find that CRB3 shRNA-mediated knockdown in these cells impairs their ability to form properly polarized acini in 3D cultures. Furthermore, they find that MCF10A cells lacking CRB3 display reduced primary ciliation frequency compared to control cells, which is in agreement with previous studies implicating CRB3 in primary cilia biogenesis. Using a combination of biochemical, molecular- and imaging approaches the authors then provide some evidence indicating that CRB3 promotes ciliogenesis by mediating Rab11-dependent recruitment of gamma-tubulin ring complex (gamma-TuRC) component GCP6 to the centrosome/ciliary base, and they also show that CRB3 itself is localized to the base of primary cilia. Finally, to assess the functional consequences of CRB3 loss on ciliary signaling function, the authors analyze the effect of CRB3 loss on Hedgehog and Wnt signaling using cell-based assays or a mouse model.

Overall, the described findings are interesting and in agreement with previous studies showing an involvement of CRB3 in epithelial cell biology, tumorigenesis and ciliogenesis. The results showing a role for CRB3 in mammary epithelial development and morphogenesis in vivo seem convincing. Although the authors provide evidence that CRB3 promotes ciliogenesis via (indirect) physical association with Rab11 and gamma-TuRC, the precise mechanism by which CRB3 promotes ciliogenesis remains to be clarified. Specific comments are as follows:

1. For all cell-based assays using shRNA to knock down CRB3, it would be desirable to perform rescue experiments to ensure that the observed phenotype of CRB3 depleted cells is specific and not due to off-target effects of the shRNA.
2. Figure 3G: it is very difficult to see that the red stained structures are primary cilia.
3. Figure 5A: it is unfortunate the authors chose not to show the original dataset (Excel file) used for generating this figure; this makes it difficult to interpret the data. It is general policy of the journal to make source data accessible to the scientific community.
4. The authors have a tendency to overinterpret their data, and not all claims put forth by the authors are fully supported by the data provided.

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Reviewer #2 (Public Review):

In this work, the authors investigate the role of CRB3 in the formation of the primary cilium both in a mouse model and in human cells. They confirm in a conditional knock-out (KO) mouse model that Crb3 is necessary for the formation of the primary cilium in mammary and renal epithelial tissues and the new-born mice exhibit classical traits of ciliopathies. In the mouse mammary gland, the absence of Crb3 induces hyperplasia and tumorigenesis and in the human mammary tumor cells MCF10A the knock-down (KD) of CBR3 impairs ciliogenesis and the formation of a lumen in 3D-cultures with less apoptosis and spindle orientation defects during cell division.

To determine the subcellular localization of CRB3 the authors have expressed exogenously a GFP-CRB3 in MCF10A and found that this tagged protein localizes in cell-cell junctions and around pericentrin, a centrosome marker while endogenous CRB3 localizes at the basal body. To dissect the molecular role of CRB3 the authors have performed proteomic analyses after a pull-down assay with the exogenous tagged-CRB3 and found that CRB3 interacts with Rab11 and is present in the endosomal recycling pathway. CRB3 KD also decreases the interactions between components of the gamma-TuRC. In addition, the authors showed that CRB3 interacts with a tagged-Rab11 by its extracellular domain and that CRB3 promotes the interaction between Rab11 and CEP290 while CRB3 KD decreased the co-localization of GCP6 with Rab11 and gamma-Tub.

Finally, the authors showed that CRB3 depletion cannot activate the Hh pathway as opposed to the Wnt pathway.

- https://doi.org/10.7554/eLife.86689.2.sa0

Author Response

The following is the authors' response to the original reviews.

> ***Reviewer # 1***
>
> *Specific comments*
>
> 1. *Figure 1: it is unclear how many mice were used for the described phenotypic analyses (panels D and E). Please clarify.*

We acknowledge that we made a mistake in failing to clearly describe the phenotypic analyses. In Figure 1D and E, we performed statistical analysis on the number of TEBs in whole mammary mounts. One mouse stained a mammary whole mount with Carmine-alum staining. Thus, "n" represents the 10 mice we analyzed. We have modified the legend of Figure 1 to " D, E. Quantification of the average number of TEBs and bifurcated TEBs in littermate Crb3fl/fl (n=10) and Crb3fl/fl;MMTV-Cre (n=10) mice at 8 weeks old" in lines 909-911.

> 1. *Figure 2: in panels B and C it is unclear how the data was quantified; the legend states "n=10", does this mean the experiment in B was done 10 times? And that 10 acini per condition were measured in panel C? In panel D a difference in 0.3% between NC and shCRB3 seems miniscule; do the authors mean 30% instead? And how many acini were counted per condition per (how many) experiments? Same applies to panels G and H, it is unclear how many cells were analyzed per (how many) experiments.*

Thanks for your suggestions. We failed to describe the details of the statistical analysis well in the experimental method. To provide a brief overview of our statistical analysis method, we took 3-4 random bright-field micrographs of each well in the chamber slide system and repeated the experiment three times. We then counted the number of acini in all micrographs (Figure 2B) and examined the diameter of all acini in each photograph, averaging the values as data (Figure 2C). We also determined the percentage of aberrant acini in each photograph, which was used as an analysis value (Figure 2D). We carefully confirmed that the vertical axis of Figure 3D was indeed mislabeled and should mean 30%, and revised the original figure. For IF analysis of the mitotic spindle orientation during lumen formation, we examined the division angle of one cell in one acinus that was mitotically dividing, 3-4 acini were randomly examined in each well in the chamber slide system, and this experiment was repeated three times (Figure 2G and H). Therefore, we have provided a detailed description of these issues in the Figure 2 legend. The revised parts are found in lines 922-924, lines 926-927, lines 929-930, and line 932.

> 1. *Figure 2: it would be desirable if authors were able to quantify the data in panels E and I.*

Thank you for your comments. According to your suggestions, we performed the quantitative analysis of Figure 2E and I, which is now presented in the new Figure 2D and H.

> 1. *For all cell-based assays using shRNA to knock down CRB3 (Fig. 2A-H; Fig. 3A-F; Fig. 4C-E; Fig. 5G-J; Fig. 6C; Fig. 7C, D; Fig. 8E-G), it would be desirable to perform rescue experiments to ensure that the observed phenotype of CRB3 depleted cells is specific and not due to off-target effects of the shRNA.*

Yes, rescue experiments involving overexpression of CRB3 in CRB3 depleted cells can accurately account for the specific phenotype as well as eliminate the off-target effects of shRNA. However, our group has long focused on the role of the cell polarity protein CRB3 in contact inhibition and tumorigenesis. Our previous studies have ruled out the off-target effects of shRNA and reported that CRB3 regulates contact inhibition and tumorigenesis through Hippo or Wnt signaling pathways (Cell Death Dis 2017;8(1):e2546, Oncogenesis 2017;6(4):e322, J Cell Mol Med 2018;22(7):3423-33). Therefore, we will pay close attention to rescue experiments to ensure experimental integrity and phenotypic specificity in our subsequent studies.

> 1. *Figure 3: how many cells were counted/measured per condition (in how many experiments) in panels B, D, H, F, G and H? In panels C and D, what is the CRB3 protein level in these cells? This is of relevance as protein overexpression per se could impinge on ciliation frequency. This question could be addressed by performing a western blot analysis with CRB3 antibody.*

We did not clearly describe the measurement and statistical analysis methods in the previous manuscript. Similarly, we took 3-4 random IF and SEM micrographs of each sample in one experiment, and this experiment was repeated three times. Subsequently, the number of ciliated cells and total cells were counted, and the proportion of ciliated cells was calculated (Figure 3B, D and F). In these figures, the cilium length of representative ciliated cells was measured in each photograph. In the knockout mouse model, we needed to find the intact mammary ductal lumen and renal tubule in IF staining of mouse mammary and renal tissue sections, with 5-6 random fields micrographs taken per slice, and the proportion of ciliated cell was measured by counting and taking the average. A total of ten mice were repeated in these experiments (Figure 3G and H). Therefore, the legend of Figure 3G and H has been partially modified and a detailed description has been added to the Figure 3 legend. The revised parts are in lines 945-946, lines 950-951, line 953.

Thank you for your suggestions that we perform a western blot analysis with CRB3 antibody in Figure 3C and D. And we have added the western blotting with CRB3 analysis in the new Supplementary Figure 3A.

> 1. *Figure 3G: it is very difficult to see that the red stained structures are primary cilia.*

Yes, the staining structure of primary cilia in mammary ductal lumen are less clear than that of individual cells and in renal tubule in Figure 3G. We used recognized acetylated tubulin and γ-tubulin to stain the primary cilia, which were clearly labeled in individual cells. However, the labeled primary cilia in renal tubule were longer length and demonstrated a more pronounced structure than those in the mammary ductal lumen. In the mammary ductal lumen of the 10 mice we analyzed, the primary cilia showed shorter length and staining structure than the others shown in Figure 3G. This difference may be due to the distinct characteristics of primary cilia in different tissues.

> 1. *Figure 4B: how many cells were analyzed in how many experiments?*

Our statistical methods for analyzing cellular experiments using IF were essentially the same. We randomly selected 3-4 IF micrographs of each sample in one experiment, and this experiment was repeated three times. Subsequently, the number of colocalization cells and total cells were counted, and the proportion of cells with pericentrin and CRB3 colocalization was calculated (Figure 4B). The detailed description has been added to the Figure 4 legend. The revised part is in lines 962-963.

> 1. *Lines 217-219: since the cells were not stained with a cilia marker, only a centrosome marker, the claim that CRB3 localizes to the base of cilia is unsubstantiated.*

Thank you for your comments. The base of cilia is the basal body, which develops from the mother centriole of the centrosome (Cancer Res. 2006;66(13): 6463-7). Firstly, we found colocalization of CRB3 and pericentrin, a centrosome marker, in MCF10A cells (Figure 4A and B). Secondly, we verified the colocalization of CRB3 with γ-tubulin, a marker of basal body in primary cilia, in confluent quiescence cells (Figure 4C and D). In addition, we found that CRB3 was localized at the base of primary cilia labeled with acetylated tubulin (Figure 4E and F). Due to the species of commercialized CRB3 antibody, we were able to indirectly claim that CRB3 localizes to the base of cilia through these experiments.

> 1. *Figure 3 and Figure 4: is it problematic to use gamma tubulin as centrosome marker if CRB3 depletion causes reduced centrosomal recruitment of gamma tubulin ring complex components? Also, in Figure S3A no gamma tubulin staining can be seen in the lower panel, why?*

Thank you for your positive comments. As is well known, γ-tubulin is a marker of the centrosome, and we found that CRB3 depletion causes reduced centrosomal recruitment of gamma tubulin ring complex components. However, Our Figure 3 was illustrated the effect of CRB3 on ciliary assembly, and Figure 4 was analyzed the localization of CRB3 in primary cilia. In some reports on ciliary assembly, the fluorescent double staining of acetylated tubulin and γ-tubulin have been used to label primary cilia, and the effect of target genes on ciliary number and assembly were analyzed by these markers (Nature. 2013;502(7470): 254-7, Cell. 2007;130(4): 678-90 and so on). Although CRB3 affects the recruitment of gamma tubulin ring complex components, it does not affect the analysis of ciliary number and localization in Figures 3 and 4.

In Figure S3A, green staining labeled with γ-tubulin could be clearly found in the lower left panel. The representative area from the left amplification may have been poorly selected, resulting in no γ-tubulin staining on the right side. We have updated the lower right panel in the new Supplementary Figure 3B.

> 1. *Figure S4A: the grouping of indicated proteins is factually wrong. For example, FBF1, SCLT1 and ODF2 are not IFT-B components, and several of the proteins indicated as localizing to the basal body also localize to (unciliated) centrioles. In contrast, CP110 is usually only found on unciliated centrioles and not mature basal bodies. Authors should consult the relevant literature and correct the figure accordingly. Alternatively, this misleading text/grouping could be removed from the figure. Furthermore, in the legend to Figure S4 there is no information provided about this quantitative analysis (how many independent experiments, which cells were analyzed etc.).*

Thank you for your helpful suggestions. We have taken your advice and removed this misleading information from the manuscript, Supplementary Figure 4A and its corresponding legend. In the legend to Supplementary Figure 4A, we have added the detailed information for this quantitative analysis in the legend. The revised legend is shown in lines 1098-1100.

> 1. *Figure S4B: how do authors know which of the bands correspond to CRB3 fusion protein?*

Based on the construction strategy of the CRB3-GFP fusion protein (Figure 6D) and its base sequence, we were able to calculate its molecular weight. Then the molecular weight of CRB3-

GFP fusion protein was verified by western blotting (Figure 6F and 7A). Meanwhile, exogenous overexpression allowed for the production of the CRB3-GFP fusion protein in large quantities. Due to these features, we could know that the band indicated by the black arrow is most likely CRB3-GFP fusion proteins. In order to check the molecular weight, we have labeled the key molecular weight markers in the new Supplementary Figure 4B.

> 1. *Lines 251-253: this seems like data overinterpretation.*

Thank you for your comments. We have revised this sentence in lines 252-254.

> 1. *Lines 260-261: the data showing perturbed gamma tubulin localization is not convincing as data was not quantified.*

According to your suggestions, we performed the quantitative analysis of Figure 4C, which is now presented in the new Figure 4E.

> 1. *Figure 5H and Figure 6C: to show that the GCP6 IP actually worked, these blots should be probed also for GCP6.*

Thank you for your good suggestions. We have added these blots probed for GCP6 in new Figure 5H and 6C.

> 1. *Figure 5I: how many cells were analyzed in how many experiments?*

Our statistical methods for analyzing cellular experiments using IF were essentially the same. We took 3-4 random IF micrographs of each sample in one experiment, and this experiment was repeated three times. The detailed description has been added to the Figure 5 legend. The revised part is in lines 992-994.

> 1. *Figure S5: it looks like GPC6 and Rab11 are localizing all over the cell, are the antibodies used for the IFMs specific for these proteins?*

After checking the specificity of these antibodies used for the IFMs, we have decided to delete the corresponding results in the Supplementary Figure 5 and their description in the original manuscript.

> 1. *Lines 43, 89, and 314-315: the claim that CRB3 directly binds Rab11 is not supported by the data. The data provided only shows that these proteins interact indirectly. To show direct interaction, yeast-2-hybrid analysis or pull-down assays with purified proteins would be required.*

Thank you for your positive comments. Since we were unable to complete the relevant experiments to demonstrate direct interaction of two proteins, we have revised our conclusions. Replace " CRB3 directly binds Rab11" with " CRB3 binds Rab11" in the manuscript.

> 1. *Figure 6G and lines 314-315: this result is surprising as it indicates GTP- and GDP-locked versions of Rab11 have the same inhibitory effect on CRB3 binding? Please comment, and also indicate how data in Figure 6G was quantified (and how many independent experiments were used for the quantification).*

We were also puzzled by the results shown in Figure 6G. Based on the western blotting bands, we suspected that there may have been some issues with the experiment. Specifically, we believed that the inefficient transfection of Flag-Rab11aWT, Flag-Rab11a[Q70L], Flag-Rab11a[S20V], and Flag-Rab11a[S25N] plasmids, as well as the insufficient amount of GFP antibody used in the co-IP experiment, led to the corresponding bands being too weak and masking the true differences.

To address this, we optimized the experimental conditions, strictly increased the experimental control, and repeated the experiment in triplicate. The new results are shown in the revised Figure 6G. The statistics from the three independent experiments revealed that CRB3b had a stronger interaction with Rab11a[Q70L] and Rab11a[S20V], while showing a weaker interaction with Rab11a[S25N], compared to Rab11aWT. As this result, we revised the original manuscript in lines 308-310 and added a detailed description to the Figure 6 legend in lines 1012-1013.

> 1. *Figure 8G: data needs to be quantified.*

Thank you for your comments. We replaced the unattractive bands in the western blotting of Figure 8G with better quality ones. The statistical analysis of the Figure 8G data is shown in Supplementary Figure 6.

> *Further minor comments*
>
> 1. *Abstract should indicate that this study describes conditional knockout of Crb3 in mouse mammary gland epithelial cells.*

This is good writing advice. We have added the relevant description in lines 40-42.

> 1. *Line 87: specify which gland (mammary?).*

We have modified to " mammary gland" in line 87.

> 1. *Line 140: sentence states that knockout of Crb3 is essential for branching morphogenesis in mammary gland development, I do not think this is correct.*

We have removed the inappropriate finding.

> 1. *Line 152: "formed more number" should be "formed more" or "formed higher number of".*

We modified "formed more number" to "formed more" in line 154.

> 1. *Lines 157-163: text and logic are difficult to follow for a non-expert.*

We have modified the logic of this paragraph, as detailed in lines 158-165.

> 1. *Figure 4A, C: figure resolution could be improved. It is difficult to see what the authors claim these figures are showing.*

The clarity of the original images in Figure 4A and C is acceptable, while the images on the right are electronically enlarged. Although there is a decrease in pixels, it can still display our findings.

> *1. Figure 7D, E: images look pixelated.*

The clarity of the original images in Figure 7D and E is acceptable using a laser confocal microscope, while the images on the right are electronically enlarged.

> *1. Line 222: unclear what authors mean by "detected a series".*

We modified "detected a series" to "some important" in line 226.

> *1. Lines 221-225: which cells were used for the analysis in Fig. S4?*

We used MCF10A cells for the analysis in Supplementary Figure 4, and modified its legend in line 1098.

> *1. Line 245: what is "cytomembrane"?*

We modified "cytomembrane" to "cell membrane" in lines 246-247.

> *1. Lines 246-250: wording is unclear/difficult to understand.*

We have modified this paragraph, as detailed in lines 248-251.

> *1. Line 273: should "regimented" be "sedimented"?*

We modified "regimented" to "sedimented" in line 274.

> *1. Line 287-288: sentence does not make sense.*

We have removed this sentence.

> *1. Figure 5A: it would be desirable to show the original dataset (Excel file) used for generating this figure.*

To maintain data integrity, we should provide the original dataset (Excel file). However, there are some unpublished data in this file that we must withhold for the time being. If needed, the corresponding author can be requested to provide the file.

> *1. Lines 298-299: wording is unclear.*

We have modified this sentence, as detailed in lines 296-298.

> *1. Lines 285-287: replace "instead of" with "but not".*

We modified "instead of" to "but not" in line 286.

> *1. For all IFMs showing merged images of the green and red channel, please also show the red and green channel separately.*

Most of our fluorescence images are presented separately for each channel in this manuscript, with only a few merged images due to space limitations. This type of presentation is commonly used in published papers.

> 1. *Lines 326 and 327: replace "bonded" with "bound".*

We have modified in lines 322-323.

> 1. *Lines 327-328 and 361-364: wording is unclear/grammatically incorrect.*

We have modified these paragraphs, as detailed in line 323 and lines 357-360.

> 1. *Line 342: what is meant by "the combination of"?*

We modified "the combination of" to "the binding of" in line 338.

> 1. *Line 365: localization of what?*

This means "subcellular localization" in lines 360-361.

> ***Reviewer # 2***
>
> *Major points*
>
> 1. *CRB3 is present in mammals as 2 isoforms, A and B, originating from alternative splicing. In this study, the authors never mention this fact and when using approaches to KO or KD CRB3A/B they are likely to deplete both isoforms which have been shown to have different C-terminal domains and functions (Fan et al., 2007). This is also important for the CRB3 antibodies used in the study since according to the material and methods section they are either against the extracellular domain common to both isoforms or the intracellular domain which is only similar in the domain close to transmembrane between the 2 isoforms. Since the antibodies used in each figure are not detailed it is impossible to know if the authors are detecting CRB3A or B or both. Please provide the information and correct for the actual isoform detected in the data and conclusions.*

Thanks for your positive comments. In mammals, CRB3 has two isoforms, CRB3a and CRB3b, distinguished by alternative splicing within the fourth exon of the CRB3 gene, which in turn produces a protein with 23 amino acid differences at the C terminus. Both CRB3a and CRB3b have mostly identical amino acid sequences, and have indistinguishable molecular weight sizes. As a result, the knockout mouse construction strategy and the design principles of RNAi sequences target both CRB3a and CRB3b. This is described in lines 100-104 and lines 149-150. Additionally, commercially available antibodies detect both CRB3a and CRB3b, as mentioned in line 123 and lines 636-637 in revised manuscript.

However, it should be noted that our CRB3 overexpression, as shown in the CRB3 structural domain in Figure 6D, refers specifically to the sequence of CRB3b. As a result, we have updated the original manuscript as well as the legends of Figures 3C, 3E, 4A, 5A, 5B, 6D-G, 7A, 7B and Supplementary Figure 2F-H, 3A, 4B, 6B to reflect this change. All instances of overexpressed CRB3 have been changed to CRB3b.

> 1. *CRB3A and B have been localized in the cilium itself (Fan et al., 2004; 2007) but in the study CRB3A/B does not enter the cilium but is localized in the basal body (figure 4). How the authors reconcile these different localizations?*

Indeed, we found that CRB3 is mainly localized at the basal body of the primary cilium, which differs from previous reports in the literature (Curr Biol. 2004;14(16):1451-61 and J Cell Biol. 2007;178(3):387-98). However, upon closer examination of one of these reports (Curr Biol. 2004;14(16):1451-61), it appears that CRB3 was actually scattered on the primary cilia, with a strong focus at the basal body. Additionally, in rat kidney collecting ducts, the localization of CRB3 on primary cilia was significantly reduced, with obvious localization at the basal body. Another study (J Cell Biol. 2007;178(3):387-98) also reported the co-localization of CRB3b and γ-tubulin in MDCK cells, which is consistent with our conclusion. We further verified the co-localization of CRB3 with the centrosome by overexpressing CRB3b in mammary epithelial cells, indicating that CRB3 mainly localizes to the basal body of the primary cilium. This information is discussed in the Discussion section of the manuscript (lines 400-410).

> 1. *The authors use GFP-CRB3A/B, it is not stated which isoform, over-expression to localize CRB3A/B in MCF10A cells (figure 4A). The levels of expression appear to be very high in the GFP panel and it is likely that the secretory pathway of the cells is clogged with GFP-CRB3A/B in transit from the ER to the plasma membrane. Thus, the colocalization with pericentrin might be due to the accumulation of ER and Golgi around the centrosome. This colocalization should be done with the endogenous CRB3A/B and with a better resolution.*

Thank you for your comments. We were also interested in the co-localization of endogenous CRB3 and centrosome proteins. However, the only commercial CRB3 antibody available is the rabbit species, and the pericentrin antibody (Abcam, ab4448) that is very useful is also the rabbit species. We had difficulty finding commercial centrosome-associated antibodies for other species. Therefore, we examined the co-localization of endogenous CRB3 with γ-tubulin in Figure 4C and combined the results with those of exogenous CRB3 to illustrate the co-localization of CRB3 with centrosomes.

> 1. *The staining for CRB3A/B in figure 4C (red) is striking with a very strong accumulation in an undefined intracellular structure and the authors do not provide any explanation for such a difference with the GFP-CRB3A/B just above.*

Thank you for your good suggestions. The immunofluorescence images of GFP-CRB3 in Figure 4a were obtained using a fluorescence microscope, while the images of endogenous CRB3 were obtained using a laser confocal microscope. The fluorescence microscope excites a fluorescent dye to emit a signal, which is amplified into a visible light signal and presents a full fluorescent signal. In Figure 4a, we can clearly see the full distribution of exogenous CRB3 in MCF10A cells, including its tight junctional localization consistent with previous reports in the literature and its co-localization with centrosomal proteins. On the other hand, laser confocal microscopy uses a laser as the light source to excite the fluorescence within the sample point by point. It employs a precision pinhole filtering technique with strong laminar imaging capabilities. In the specific analysis of endogenous CRB3 co-localization studies with centrosomes and primary cilium, signals at tight junctions must be excluded. Therefore, Figure 4c represents the fluorescence signal at the level of intracellular CRB3 co-localization with γ-tubulin. The two methods use different detection means and techniques, and are not directly comparable.

> 1. *The staining in figure 4E is also different from those shown in figure 4F in which the CRB3A/B staining is right at the base of the axoneme while it is not the case in figure 4E where we can see a red dot close to but not right at the base of the axoneme.*

Thank you for your comments. The new Figure 4F displays the localization relationship between CRB3 and primary cilium, analyzed using laser confocal microscopy. With the unique single-level detection function of this microscope, the problem of level selection may cause the red dots to appear close to, rather than right at the basal body of the primary cilium. However, the new Figure 4G, based on the use of 3D reconstruction scanning technique, clearly demonstrates the localization of CRB3 at the basal body of the primary cilium under the same cells and conditions.

> 1. *The authors claim that CRB3A/B interacts directly with Rab11 but they only show co-immunoprecipitation experiments from cell lysates which do not support direct interactions. The only way to show a direct interaction is to produce both proteins in vitro. Thus, the term direct interaction should be removed.*

Thank you for your positive comments. Since we were unable to complete the relevant experiments to demonstrate direct interaction of two proteins, we have revised our conclusions. Replace " CRB3 directly binds Rab11" with " CRB3 binds Rab11" in the manuscript.

> 1. *In addition, the authors claim (Line 251/252) that Rab11 is necessary for the transport of CRB3A/B but they should KD Rab11 to show this.*

Thank you for your good suggestions. It is essential to observe CRB3 trafficking after knockdown Rab11. However, in Figure 5C, we used the endocytosis inhibitor dynasore, which also inhibits Rab11-positive endosomes. This result shows that dynasore can significantly inhibit CRB3 trafficking in MCF10A cells. We believe that this experiment partially demonstrates that inhibiting Rab11 function can affect CRB3 trafficking.

> 1. *The domain of CRB3A/B that is necessary for the interaction with Rab11 is the N-terminal part of the extracellular domain. This domain is thus inside the transport vesicles and not accessible from the cytoplasm. Given that Rab11 is a cytoplasmic protein, how the 2 proteins could interact across the membrane? The authors do not even discuss this essential point for their hypothesis.*

Thank you for your positive comments. As shown in the schematic model in Figure 9, we believe that when cells form tight junctions, CRB3 is primarily located on the cell membrane. Subsequently, endosomes are involved in the intracellular degradation process of CRB3 on the cell membrane. Intracellular CRB3 can bind to Rab11 through the extracellular domain, which in turn participates in primary cilia assembly. We have made detailed modifications to lines 418-421.

> 1. *Figures are not numbered.*

Thank you for your comments. We have updated the numbers in the original manuscript as well as the legends of Figures 1D, 1E, 2B, 2D, 2F, 2G, 3B, 3D, 3F-H, 4B, 4E, 5I, 6, 8G and Supplementary Figure 1E, 2, 3C, 4A, 5B, 6.

> *Minor points*

> 1. *The authors cite several studies showing that a down regulation of CRB3A/B in human cells promotes cancer but other studies show the contrary: Lin et al., 2015 for example. Please discuss these discrepancies.*

Thanks for your good suggestion. We have included additional studies with contrasting results in the discussion section, specifically in lines 378-380.

> 1. *Line 98: "exhibit smaller" smaller than what?*

We modified "exhibit smaller" to "exhibit smaller size" in line 97.

> 1. *Line 152: "form more number, ..." ???*

We modified "formed more number" to "formed more" in line 154.

> 1. *Line 180: "Compared with the control, the number of cells with primary cilium was significantly increased ». To me it is the contrary! This part is not clear at all. Please rewrite.*

We have revised the sentence in lines 183-185.

> 1. *Authors should check and review extensively for improvements to the use of English.*

Thanks for your good writing advice. We have carefully reviewed and revised the entire manuscript to improve its readability.